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Nurse-led vs. usual-care on cardiovascular risk factors for patients with type 2 diabetes: a systematic review and meta-analysis

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Running Title: Nurse-led care for patients with T2DM

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Abstract:

Objects: Our study aims to systematically evaluate nurse-led care's effectiveness on cardiovascular risk factors among individuals with type 2 diabetes mellitus.

Design: Systematic review and meta-analysis.

Methods: The electronic databases PubMed, EMBASE, CINAHL and Cochrane Library databases was conducted for randomized controlled trials of nurse-led care for individuals with type 2 diabetes mellitus published in English up to April 27, 2021. Random effects models and inverse variance weighting were used to calculate relative risks with 95% confidence intervals.

Results: As many as 13 articles were included in the meta-analysis, for a total of 3475 participants. Pooled analysis showed there were no differences in low-density lipoprotein cholesterol (LDL-c) (WMD=-0.02mmol/L; 95% CI -0.12, 0.09; p=0.76), systolic blood pressure (SBP) (WMD=0.34; 95% CI -2.20, 2.89; p=0.79) and body mass index (BMI) (WMD=-0.27; 95% CI -0.68–0.14; p=0.19) between the nurse-led and control groups. However, the nurse-led management program significantly decreased the glycosylated hemoglobin (HbA1c) (WMD=-0.58, 95% CI = -0.73 to -0.43; P<0.001) for patients with T2DM.

Conclusion: Nurse-led care is an effective and accessible intervention that could improve glycemic control among individuals with type 2 diabetes mellitus.

Keywords: cardiovascular; meta-analysis; nurse-led care; systematic review; type 2 diabetes mellitus.

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Strengths and limitations of the study

1 Nurses play pivotal roles in diabetes care, and nurse-led interventions may have benefits on promoting early complications detection.

2 In this meta-analysis, nurse-led care is an effective and accessible intervention that could improve the glycemic control among individuals with type 2 diabetes mellitus.

3 Compared with telephone interventions, face-to-face interventions were more effective in decreasing HbA1c levels.

4 Future research should focus on strategies for sustaining glycaemic treatment among face-to-face interventions effects in the long term.

5 Compared with other nurse-led RCTs, the sample size of studies we included was small, more RCTs are needed in future to help us draw a solid conclusion.

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1. Introduction

The increasing incidence of Type 2 diabetes (T2DM) has significant implications for healthcare management. According to the World Health Organization (WHO), there are currently more than 463 million T2DM globally[1]. T2DM is associated with several complications[2], in which cardiovascular complications are the main leading of mortality[3-6]. The incidence of myocardial infarction and sudden death in T2DM is the same as that of patients with coronary heart disease[7, 8]. Therefore, it is crucial to control the cardiovascular risk factors for patients with T2DM.

Previous studies[9, 10] have shown that most patients with T2DM were under the management of physicians, which has led to a gradual increase in pressure workload for physicians. Hence, many investigators extended the role of nurses as case managers[11]. To address evolving clinical requirements, nurses, may pick a certain number of patients that can be independently within advanced practice, and thus relieving the stress within medical clinics[12]. There is mounting evidences[13-17] indicated that nurse-led diabetes management program could provide more comprehensive care, including guidance to diet, symptom management, lifestyle changes, psychological support, and diabetic education, which can prevent or delay the complications of diabetes. However, disputable conclusions remained existing in different studies. For example, a study conducted by Tang PC and colleagues[18] showed a great reduction in glycosylated hemoglobin (HbA1c) with an intervention period of 6 months, but no difference for 12 months (P=0.133). Morgan MA, et al.[19]

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found that both experimental groups showed a remarkable decrease in HbA1c(p=0.049). Vos RC, et al.[20] reported no significant difference between the nurse-led and control groups. Furthermore, several trials published recently need updated evidence synthesis. Therefore, we conducted a systematic review and meta-analysis to evaluate the impact of a nurse-led interventions program versus usual-care on patients with T2DM and provide trustworthy practice guidelines on newly released evidence.

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2. Methods

This systematic review was registered on PROSPERO (registering number: CRD42021248275), and conducted following the PRISMA Recommendations project[21]. We devised research questions using the PICO framework for systematic review: for patients with T2DM (Participants), nurse-led diabetes education (Intervention), cardiovascular risk factors outcomes (Outcomes) for the intervention group than the usual care group (Comparison) in randomized controlled trials (RCTs) (Type of studies). We defined nurse-led care[22] as that a nurse plays a central role during T2DM management, follows up patients, monitors blood tests results, and provides continuous education.

2.1 Data sources and searches

The literature search was conducted in PubMed, Embase, CINAHL, and the Cochrane Central Register of Controlled Trials until April 27, 2021. We used the following Mesh terms in combination using AND or OR: "Diabetes Mellitus, Type 2", "noninsulin-dependent diabetes mellitus", "Practice Patterns, Nurses", "Nursing Practice". We limited the search language to English. We also searched the reference list of previous reviews related to this systematic review and contacted the study authors for more data when required (The search strategy is detailed in supplementary appendix 1).

2.2 Study selection

To be included in this review, studies had to meet all the following criteria: (1)

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randomized controlled trials; (2) had at least one of cardiovascular risk factors for diabetes as an outcome; (3) written in English; (4) and had more than 6 months of follow-up. Trials were excluded (1) if the intervention was delivered by other healthcare providers; (2) duplicate publications.

2.3 Outcomes

The outcome included any cardiovascular risk factor including HbA1c, systolic blood pressure (SBP), low-density lipoprotein cholesterol (LDL-c), Body Mass Index (BMI), and a combination of all four.

2.4 Data extraction and quality assessment

Two reviewers (JZ and XZ) worked independently to scan and evaluate full text for eligible studies using Note-Express software. The third reviewers (CL and YD) resolved disagreements by discussion. Relevant data included study authors, publication year, country, delivery setting, age, sample size, length of follow-up, description of interventions, outcomes, and the data checked for accuracy, which two reviewers extracted. All data were collated, imported to a statistical evaluation program (Excel sheet), and prepared for analysis.

The Cochrane risk-of-bias tool[23, 24] were used to evaluate the bias of randomized studies. Allocation concealment, random sequence generation, blinding of participants, personnel and outcome assessors, selective reporting of outcomes, and incomplete

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outcome data were carefully evaluated.

2.5 Data synthesis and analysis

For the statistical analysis[25], the Review Manager (Version 5.4.0) software was used. We have calculated mean differences (MDs) with 95% confidence intervals (CIs) and pooled values for rates presented as rate ratios. Heterogeneity was evaluated by $\chi 2$ test and corresponding P-value to assess the dispersion of the actual effect among the included studies. I^2>50% was considered high heterogeneity (the random-effects model was used), and values lower than 50% to indicate low heterogeneity (fixed effects model was used). To further explore the effect of different delivery settings, we conducted a subgroup analysis of the factors that may lead to heterogeneity. A sensitivity analysis was performed to explore the effect of each study on the overall pooled estimate. We used funnel plot to detect the publications bias. P value<0.05 indicated a statistically significant difference.

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3. Results

3.1 Searching process

A total of 2,229 articles were retrieved. After the removal of duplicates, 1311 articles were retained for abstract and title screening. Initial screening excluded 1,844 articles; only 171 articles remained for full text screening. Further, 158 articles were excluded due to various reasons such as non-nurse-led (n=57), non-eligible publication types (n =21), less than 6 months of follow-up (n = 14), conference abstract (n=14), non-English articles (n=10), not RCTs (n=9), duplicate reporting (n = 8), unrelated to the pre-decided outcomes (n=9), study protocol (n=9), ongoing study(n=4), dissertation (n=1), not relevant (n=1), and no sufficient data (n=6). So, only 13 studies were eventually left for this analysis (Fig.1).

3.1 Characteristics of the studies included.

Included trials and patient characteristics at baseline were shown in Table 1. Of the 13 studies, two were from the United States of America(USA), and the others were from Australia, Netherlands, South Korea, New Zealand, Dutch, Iran, Sri Lank, Northern Sweden, China, Italian, Japan, Belgium, Canada, and the United Kingdom(UK). Care settings and intervention modalities were conducted including remote delivery (Kim HS, et al,2007; Weinberger M, et al,1995; Tang PC, et al,2013; Odnoletkova I, et al,2016)[18, 26-28], clinical care (Vos RC, et al,2019; Azami G, et al,2018; Jayasuriya R, et al,2015; Franciosi M, et al,2011; Jansink R, et al,2013; Al Lenjawi B, et al,2016)[14, 20, 29-32] and primary care (Hörnsten A, et al,2004; Guo Z, et al,2019;

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Morgan MA, et al,2013)[19, 33, 34].

3.2 Nurse-led care program of Included Trials

All studies clearly described programs in nurse-led intervention, such as treatment of case group, intervention methods, practice recruitment, and length of follow-up. The interventions were distributed including structured diabetes education (Hörnsten A, et al, 2005; Al Lenjawi B, et al, 2016; Franciosi M, et al,2011; Jansink R, et al,2013)[30-33], trueblue collaborative care (Morgan MA, et al, 2013)[19], tele-coaching (Odnoletkova I, et al, 2016; Weinberger M, et al, 1995)[27, 28], online disease management (Tang PC, et al,2013)[18], short message service (SMS) program (Kim HS, et al, 2007)[26], beyond good intentions (BGI) program (Vos RC, et al, 2019)[20] and the diabetes self-management education/diabetes self-management (DSME/DSM) program (Azami G, et al, 2018; Jayasuriya R, et al, 2015; Guo Z, et al,2019)[14, 29, 34]. All studies reported that nurses played a leading role in the entire intervention process. The interventions components consisted of the face-to-face session, multiple group sessions, and online consulting service. All selected patients received exercise management, blood glucose monitoring, psychological adjustment, drug therapy, and insulin injection technology. In two studies (Azami G, et al, 2018; Guo Z, et al,2019)[14, 34], patients received multidisciplinary team management including endocrinologists, nutritionists, nurses, and pharmacists, in which the actual intervention to be carried out by a nurse. In 11 studies, patients received education from different types of nurses (Hörnsten A, et al, 2005; Al Lenjawi B, et al, 2016; Franciosi M, et al,2011; Jansink R,

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et al,2013; Morgan MA, et al, 2013; Odnoletkova I, et al, 2016; Weinberger M, et al, 1995; Tang PC, et al,2013; kim HS, et al, 2007; Vos RC, et al, 2019; Jayasuriya R, et al, 2015)[18-20, 26-33] such as practice nurses (PNs), primary care nurses, nurse care managers (NCMs), and community nurses, et al.

3.3 Methodological quality assessment

The eligible studies included 13 RCTs, and the seven items of e Cochrane Handbook 5.1.0 assessing the quality of these included studies, shown in supplementalry appendix 2.

3.4 Meta-analysis

3.4.1 HbA1c.

Twelve trials had investigated the effects of nursing intervention on HbA1c(Fig.2(A)). Nurse-led intervention lowered HbA1c compared with usual care (WMD=-0.58; 95% CI: -0.73 to-0.43); Heterogeneity was high indicating large variation between trial results (I^2 = 84%; p<0.0001). Sensitivity analysis by individual trial exclusion showed that two trials (Azami G, et al,2018, Jansink R, et al, 2009)[14, 31] have the greatest effect on the mean difference and heterogeneity. After excluding the above two trials the size effect and heterogeneity for HbA1c were significantly reduced (WMD=-0.54; 95% CI: 0.64, 0.54, p < 0.00001; I^2 = 54%, p<0.00001).

Subgroup analysis was performed based on different delivery setting(Fig.3). Studies

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in clinical setting (WMD=-0.70; 95% CI -1.10, -0.30, p = 0.0005) and primary care setting (WMD=-0.67; 95%Cl-0.97, -0.39, p<0.00001) showed a greater mean reduction in HbA1c compared to trials of remote delivery (WMD=-0.45; 95% CI -0.67, -0.22, p < 0.0001).

3.4.2 BMI.

Nine trials had investigated the effects of nursing intervention on BMI(Fig.2(B)). In the pooled analysis, compared to nurse lead interventions, no significant mean difference for BMI was observed for usual care (WMD=-0.27; 95% CI -0.68–0.14; p=0.19). However, only two studies were conducted by Franciosi M, et al.[30] and Odnoletkova I, et al.[28] showed a significant difference between the groups over 6 months and 18 months (p = 0.03, P=0.001, P=0.023, respectively). BMI in other studies (Jansink R, et al, 2013; Al Lenjawi B, et al, 2016; Morgan MA, et al, 2013; Vos RC, et al, 2019; Azami G, et al, 2018; Jayasuriya R, et al, 2015; Hörnsten A, et al, 2005;)[14, 19, 20, 29, 31-33] had no significant difference between the groups (p>0.05).

3.4.3 LDL-c.

There was no significant mean reduction in LDL-c in 8 trials for patients in the two group (WMD=-0.02mmol/L; CI 95% -0.12, 0.09; p = 0.76) and heterogeneity was high (I^2 = 72%, p =0.0009). A study by Jayasuriya R, et al.[29] showed that a significant reduction in LDL-c in the "usual care" group but not in the intervention group (P = 0.082). Tang PC, et al.[18] reported that the invention group had

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significantly better management of LDL-c at 12 months (-6.1mg/dl VS 0.0mg/dl, P=0.001) than that in usual care. Vos RC and colleagues (2019)[20] observed a decrease in LDL-c (p = 0.01) in the intervention group compared to controls. Other studies showed there was no statistically significant difference in LDL-c between the two groups (P>0.05) (Fig.2(C)).

3.4.4 SBP.

Of the 13 included studies, 9 investigated SBP levels as a primary outcome in T2DM (Tang PC, et al,2013; Franciosi M, et al,2011; Jansink R, et al, 2013; Odnoletkova I, et al,2016; Al Lenjawi B, et al,2016; Morgan MA, et al,2013; Vos RC, et al,2019; Azami G, et al,2018; and Jayasuriya R, et al,2015)[14, 18-20, 28-32]. No significant mean reductions were found for SBP(WMD=0.34; 95% CI -2.20, 2.89; p=0.79)in the nurse-led intervention group compared with patients who received usual care. Among these studies, 5 studies (Tang PC, et al,2013; Jansink R, et al, 2013; Morgan MA, et al,2013; Azami G, et al,2018; and Jayasuriya R, et al,2015)[14, 18, 19, 29, 31] reported that there was a slight improvement in SBP in the follows-up months. Other studies (Franciosi M, et al,2011; Odnoletkova I, et al,2016; Al Lenjawi B, et al,2016; and Vos RC, et al,2019)[20, 28, 30, 32] found that there was no significant difference in SBP between the groups (p>0.05) (Fig.2(D)).

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4. Discussion

This meta-analysis evaluate nurse-led multiple interventions that targeted a wide range of cardiovascular risk factors in patients with T2DM. Another meta-analysis[35, 36] have investigated all cardiovascular risk factors but included type 1 diabetes. The articles we included were from both developed and developing countries, and the participants came from different ethnicities, which could more fully reflect the T2DM from all over the world. The current study indicated that implementing a nurse-led care could improve mean HbA1c levels in patients with T2DM. However, no improvements were observed in mean differences in BMI, LDL-c**,** and SBP.

HbA1c has been considered a vital element in the therapy of T2DM. Our study found nurse-led care can improve patients' glycemic control, which is consistent with the previous meta-analyses (Daly B et al,2017; Tshiananga JK et al,2012; Wang Q, et al,2019; Yu-Mei Chen D, et al, 2019)[35-38]. The United Kingdom Prospective Diabetes Study(UKPDS) reported that for every 1% lower in HbA1c, microvascular complications would be reduced by 30 percent [39]. In the current study, the overall effect size was 0.58 of a standard deviation improvement in HbA1c compared to usual care, which means nurse-led diabetes care could alleviate glycaemic profiles and improving microvascular complications. Subgroup analyses revealed that the effect size of nurse-led programs on glycemic control was significantly influenced by delivery setting. Compared with telephone interventions, face-to-face interventions were more effective in decreasing HbA1c levels, at least at six months. The evidence

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that face-to-face interventions decreasing HbA1c levels in the short term is encouraging but not sufficient to allow a definitive conclusion presently. Future research should focus on strategies for sustaining glycaemic treatment among face-to-face interventions effects in the long term.

The lack of an effect from nurse-led interventions on BMI in our study was consistent with the study of Daly et al.[35]. Lean et al.[40] had shown that weight management takes at least 20-30 months, but most of the trials included in the current study managed patients for 6-12months, in which "weight rebounds" could be prone to occur at the beginning of weight management[41, 42]. Evidence suggests that weight management is a lifelong behavior and requires tailored and a strong focus on patient skills training and self-management education over the patient's lifetime. In contrast to the finding of Massimi et al.[43], we found that nurse-led RCTs have little effect on SBP and LDL-c. It might because nurses in most countries are unable to prescribe and increase titration treatments, and they rely on physicians to play this role, which may hinder the development of nursing roles[44-46].

Due to the new developments in the complexity of the disease, T2DM requires long-term care rather than intermittent treatment[47, 48]. This highlights the necessity of nursing work, and nurses can provide guidance and advice to patients in terms of disease management, which could increase the patient's knowledge and confidence. During the special period of the global novel coronavirus pneumonia epidemic, nurses

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from all over the world are actively responding to the country's call in the fight against the epidemic[49]. Before that, nurses perform a neglected human resource in China[50]. In order to improve understanding and respect for the nurse, we need to clearly define the role of nurses. We should also determine the service needs of diabetes nurses and the standards of future careers to reflect the skills and academic achievements of entry and role development.

The limitations of this study are as follows. First, there was significant heterogeneity in all conducted meta-analyses. The cultural differences may be the potential reason for the heterogeneity in 13 trials from 12 countries. Second, compared with other nurse-led RCTs, the sample size of studies we included was small, and the follow-up time of most studies is not sufficient. More RCTs are needed in future to help us draw a solid conclusion. Finally, Type 1 diabetes mellitus is recently increasingly appeared in young adults, more studies are needed to examine this population.

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5. Conclusion

This review demonstrated that nurse-led interventions can alleviate glycaemic profiles in patients with T2DM. Nurse-led diabetes education is an indispensable and important part of disease management, which makes patients and their families fully understand the hazards of the disease and increase the awareness of active participation.

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Author Contributions

The study concept and design were framed by CL and YD. JZ and XZ worked independently, scanned the titles, abstracts and evaluated full text for eligible studies using Note-Express software. CL and YD resolved disagreements by discussion. JZ conducted the statistical data analysis and drafted the manuscript. XZ and DM contributed to discussion and revision. CL and YD contributed to revision. All authors read and approved the final manuscript.

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Competing interests

None declared.

Data Availability Statement

All data from systematic review available in paper and supplementary material.

Provenance and peer review

Not commissioned; externally peer reviewed.

Ethics statement

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The proposed study does not involve collection of primary data. Accordingly, no ethical approval is required.

Patient and Public Involvement statement

The proposed study does not involve patients and the public in the design, or conduct, or reporting, or dissemination plans. Accordingly, no Patient and Public Involvement statement is required.

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Table Legends

Table 1. Characteristics of included studies.

Figure Legends

Figure 1. PRISMA flow diagram showing study selection.

Figure 2. Meta-analysis.

Figure 3. Subgroup analysis of HbA1c.

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Study author	Year	Country	Sample size (n)	Delivery setting	follow-up(month)	Mean age of usual care / nurse-led (year)	Intervention team	Treatment of case group	Treatment of control group
Kim HS, et al	2007	South Korea	51	Remote delivery	6	47.5±9.1 /46.8±8.8	Nurse case managers; Physicians	SMS	Usual care
Weinberger M, et al	1995	USA	275	Remote delivery	12	63.2±8.3 /63.9±8.6	Nurse case managers; physicians	Telephone intervention	Usual care
Vos RC, et al	2019	Dutch	108	Clinial setting	30	61.7±7.4 /62.9±8.3	Practice nurses ; physicians	DSME	Usual care
Azami G, et al	2018	Iran	142	Clinial setting	6	53.49±10.98 /55.09±10.16	A multidisciplinary team including endocrinologists, nutritionists, nurses, and pharmacists	DSME+MI	Usual care

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Jayasuriya R, et al	2015	Sri Lank	60	Clinial setting	6	51.4±7.1 /51.5±7.5	Nurse case managers; physicians	DSM	Usual care
Hörnsten A, et al	2004	Northern Sweden	104	Primary care setting	12	63.4±9.1 /63.6±9.3	Diabetes nurses; physicians	Diabetes education	Usual care
Guo Z, et al	2019	China	171	Primary care setting	12	64.35±7.07 /63.12±8.02	A multidisciplinary team including community nurses, community doctors, clinical nursing specialist, diabetes specialists, nutritionist	NLTM	Usual care
Tang PC, et al	2013	USA	415	Remote delivery	12	53.5±10.2 /54.0±10.7	Nurse care managers; pharmacist; registered dietician	EMPOWER-D	Usual care

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Franciosi M, et al	2011	Italian	62	Clinial setting	6	48.7±0.6 /48.9±0.5	Diabetes nurses; physicians	disease management	Usual care
Jansink R, et al	2013	Netherlands	940	Clinial setting	14	63.9±9.8 /64.1±8.9	Primary care nurses; physicians	lifestyle counselling	Usual care
Odnoletkova I, et al	2016	Belgium	287	Remote delivery	6	62.4±8.9 /63.8±8.7	Nurse case managers; physicians	COACH	Usual care
Al Lenjawi B, et al	2016	UK	460	Clinial setting	12	55±9.7 /52±8.9	Nurse educators ; physicians	Theory-based educational	Usual care
Morgan MA, et al	2013	Australian	400	Primary care setting	12	67.6±11.2 /68.0±11.7	Practice nurses; physicians	TrueBlue model of collaborative care	Usual care

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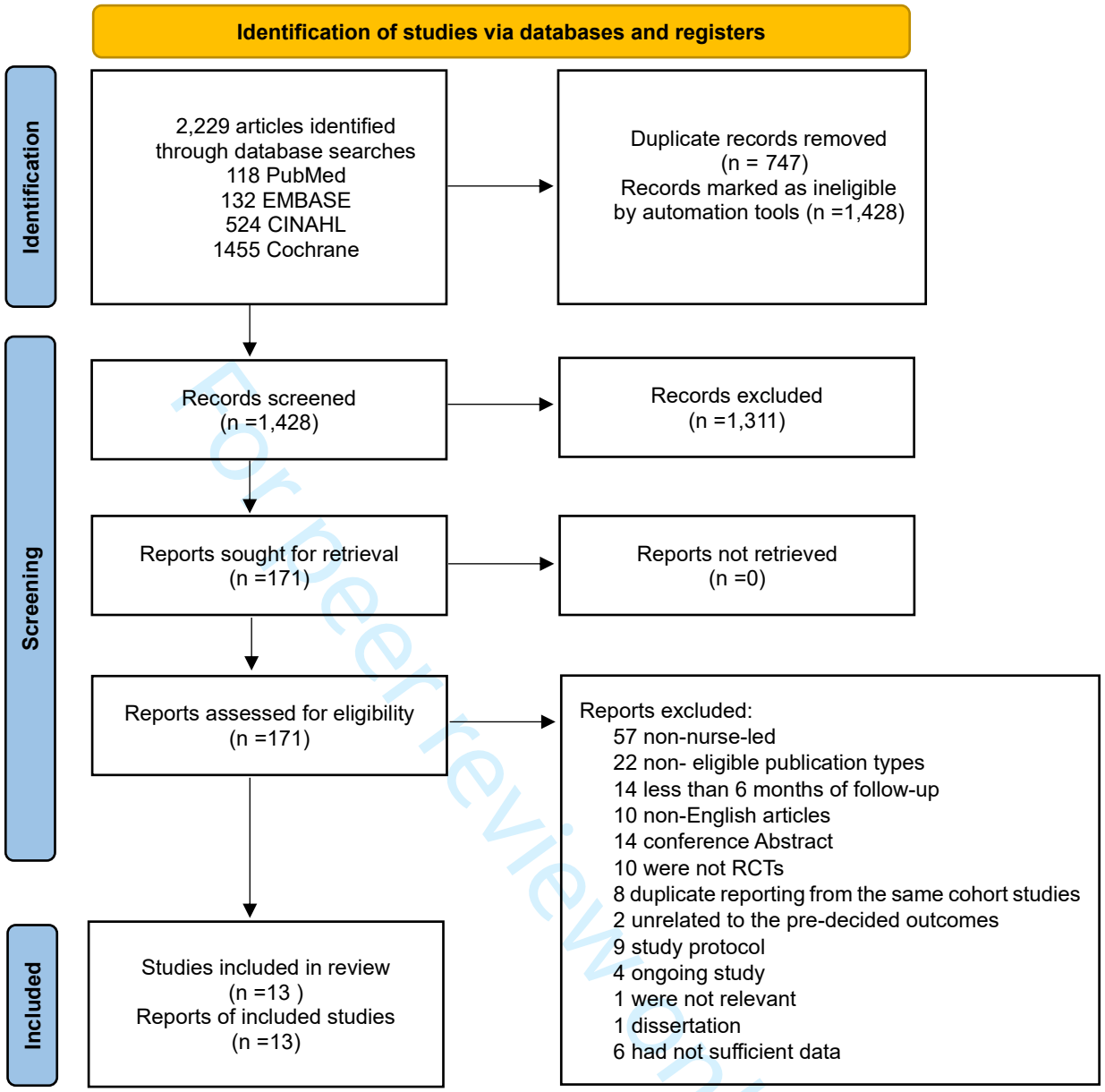
SMS : short message service; DSME: diabetes self-management education; MI : Motivational Interviewing; DSM: diabete self-management; NLTM : nurse-led team management; EMPOWER-D : Motivating Patients Online With Enhanced Resources for Diabetes; COACH: Tele-education.

Table.1 Characteristics of included studies

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Fig.1 PRISMA flow diagram showing study selection.



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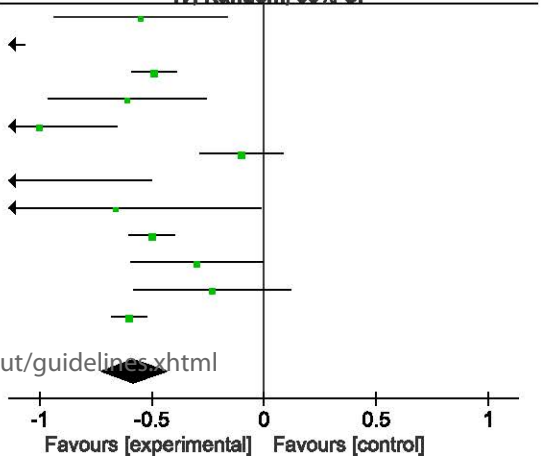
Fig 2. meta-analysis

(A) The forest plot of nurse-led clinic in HbA1c level

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	7.87	1.38	109	8.42	1.99	181	6.8%	-0.55 [-0.94, -0.16]
Azami G, et al 2018	7.9	0.93	71	9.3	1.1	71	7.8%	-1.40 [-1.74, -1.06]
Franciosi M, et al 2011	6.71	0.1	46	7.2	0.2	16	12.2%	-0.49 [-0.59, -0.39]
Guo Z, et al 2019	7.18	1	86	7.79	1.33	85	7.5%	-0.61 [-0.96, -0.26]
Hörnsten A, et al 2005	5.4	0.7	44	6.4	1.1	60	7.6%	-1.00 [-1.35, -0.65]
Jansink R, et al 2013	7.3	0.7	129	7.4	1	197	10.8%	-0.10 [-0.28, 0.08]
Jayasuriya R, et al 2015	7	1.2	28	8.3	1.7	25	2.7%	-1.30 [-2.10, -0.50]
Kim HS, et al 2007	7.04	1.39	25	7.7	0.9	26	3.7%	-0.66 [-1.31, -0.01]
Morgan MA, et al 2013	6.9	0.26	89	7.4	0.36	67	12.3%	-0.50 [-0.60, -0.40]
Odnoletkova I, et al 2016	7.4	1	108	7.7	1.2	108	8.6%	-0.30 [-0.59, -0.01]
Tang PC, et al 2013	8.1	1.68	186	8.33	1.81	193	7.5%	-0.23 [-0.58, 0.12]
Weinberger M, et al 1995	10.5	0.2	188	11.1	0.3	63	12.5%	-0.60 [-0.68, -0.52]
Total (95% CI)			**1109**			**1092**	**100.0%**	**-0.58 [-0.73, -0.43]**

Heterogeneity: Tau² = 0.04; Chi² = 66.75, df = 11 (P < 0.00001); I² = 84%

Test for overall effect: Z = 7.67 (P < 0.00001)



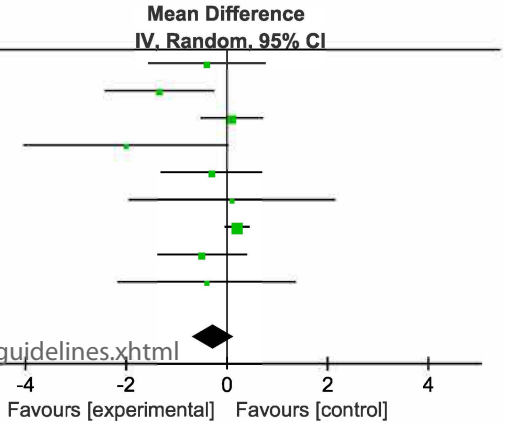
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(B) the forest plot of nurse-led clinic in BMI level

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	32.4	4.87	109	32.8	4.97	181	8.9%	-0.40 [-1.57, 0.77]
Azami G, et al 2018	28.55	3.2	71	29.89	3.4	71	9.8%	-1.34 [-2.43, -0.25]
Franciosi M, et al 2011	30.2	0.7	46	30.1	1.2	16	18.2%	0.10 [-0.52, 0.72]
Hörnsten A, et al 2005	28.7	4.6	44	30.7	6	60	3.6%	-2.00 [-4.04, 0.04]
Jansink R, et al 2013	30.2	4	106	30.5	4.6	179	10.7%	-0.30 [-1.32, 0.72]
Jayasuriya R, et al 2015	26.5	3.7	28	26.4	3.9	25	3.5%	0.10 [-1.95, 2.15]
Morgan MA, et al 2013	31.2	1	162	31	1	103	28.2%	0.20 [-0.05, 0.45]
Odnoletkova I, et al 2016	29.9	5	238	30.4	5.1	246	12.5%	-0.50 [-1.40, 0.40]
Vos RC, et al 2019	29.2	4.8	55	29.6	4.5	51	4.6%	-0.40 [-2.17, 1.37]
Total (95% CI)			**859**			**932**	**100.0%**	**-0.27 [-0.68, 0.14]**

Heterogeneity: Tau² = 0.14; Chi² = 14.28, df = 8 (P = 0.07); I² = 44%

Test for overall effect: Z = 1.31 (P = 0.19)



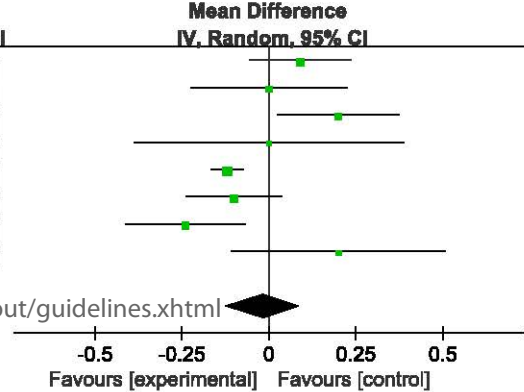
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(C) the forest plot of nurse-led clinic in LDL-c level

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	2.98	0.63	109	2.89	0.6	181	14.8%	0.09 [-0.06, 0.24]
Azami G, et al 2018	2.43	0.72	71	2.43	0.65	71	10.7%	0.00 [-0.23, 0.23]
Jansink R, et al 2013	2.6	0.8	106	2.4	0.6	178	13.2%	0.20 [0.02, 0.38]
Jayasuriya R, et al 2015	2.22	0.65	28	2.22	0.78	25	5.4%	0.00 [-0.39, 0.39]
Morgan MA, et al 2013	2.17	0.14	154	2.29	0.2	86	19.9%	-0.12 [-0.17, -0.07]
Odnoletkova I, et al 2016	2.15	0.75	237	2.25	0.78	239	15.3%	-0.10 [-0.24, 0.04]
Tang PC, et al 2013	2.3	0.87	183	2.54	0.84	189	13.3%	-0.24 [-0.41, -0.07]
Vos RC, et al 2019	2.5	0.9	54	2.3	0.7	50	7.5%	0.20 [-0.11, 0.51]
Total (95% CI)			942			1019	100.0%	-0.02 [-0.12, 0.08]

Heterogeneity: Tau² = 0.01; Chi² = 24.68, df = 7 (P = 0.0009); I² = 72%

Test for overall effect: Z = 0.30 (P = 0.76)



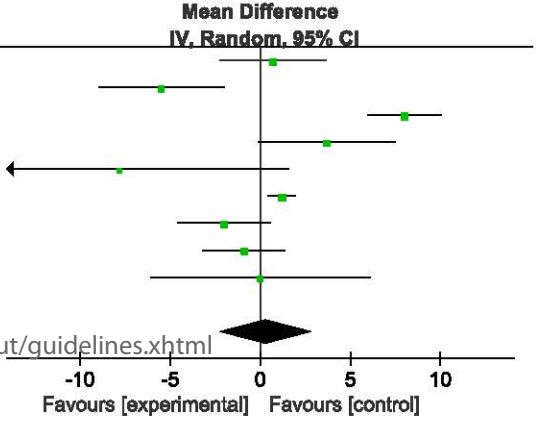
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(D) the forest plot of nurse-led clinic in SBP level

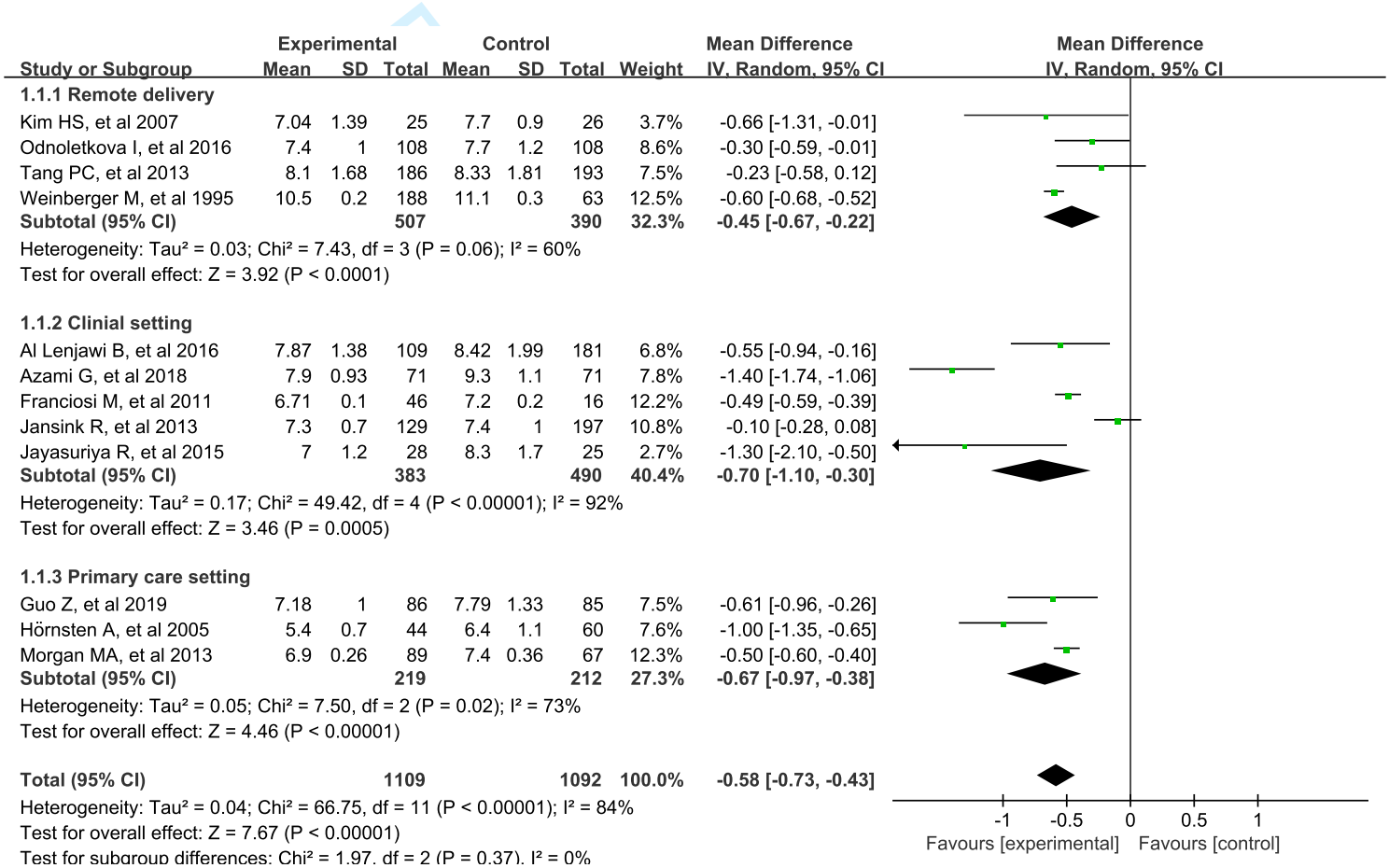
Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	136.1	12.53	109	135.4	12.3	181	12.1%	0.70 [-2.26, 3.66]
Azarni G, et al 2018	127.8	9	71	133.3	12	71	11.3%	-5.50 [-8.99, -2.01]
Franciosi M, et al 2011	133	2	46	125	4	16	13.2%	8.00 [5.96, 10.04]
Jansink R, et al 2013	141.5	17	120	137.8	15.8	185	10.9%	3.70 [-0.10, 7.50]
Jayasuriya R, et al 2015	124.1	15.9	28	131.9	18.6	25	4.9%	-7.80 [-17.17, 1.57]
Morgan MA, et al 2013	132.4	2.8	161	131.2	3.4	112	14.2%	1.20 [0.44, 1.96]
Odnoletkova I, et al 2016	128	14	241	130	15	247	12.6%	-2.00 [-4.57, 0.57]
Tang PC, et al 2013	119.9	11.4	189	120.8	11.5	192	12.9%	-0.90 [-3.20, 1.40]
Vos RC, et al 2019	135	17	55	135	15	51	7.9%	0.00 [-6.09, 6.09]
Total (95% CI)			**1020**			**1080**	**100.0%**	**0.34 [-2.20, 2.89]**

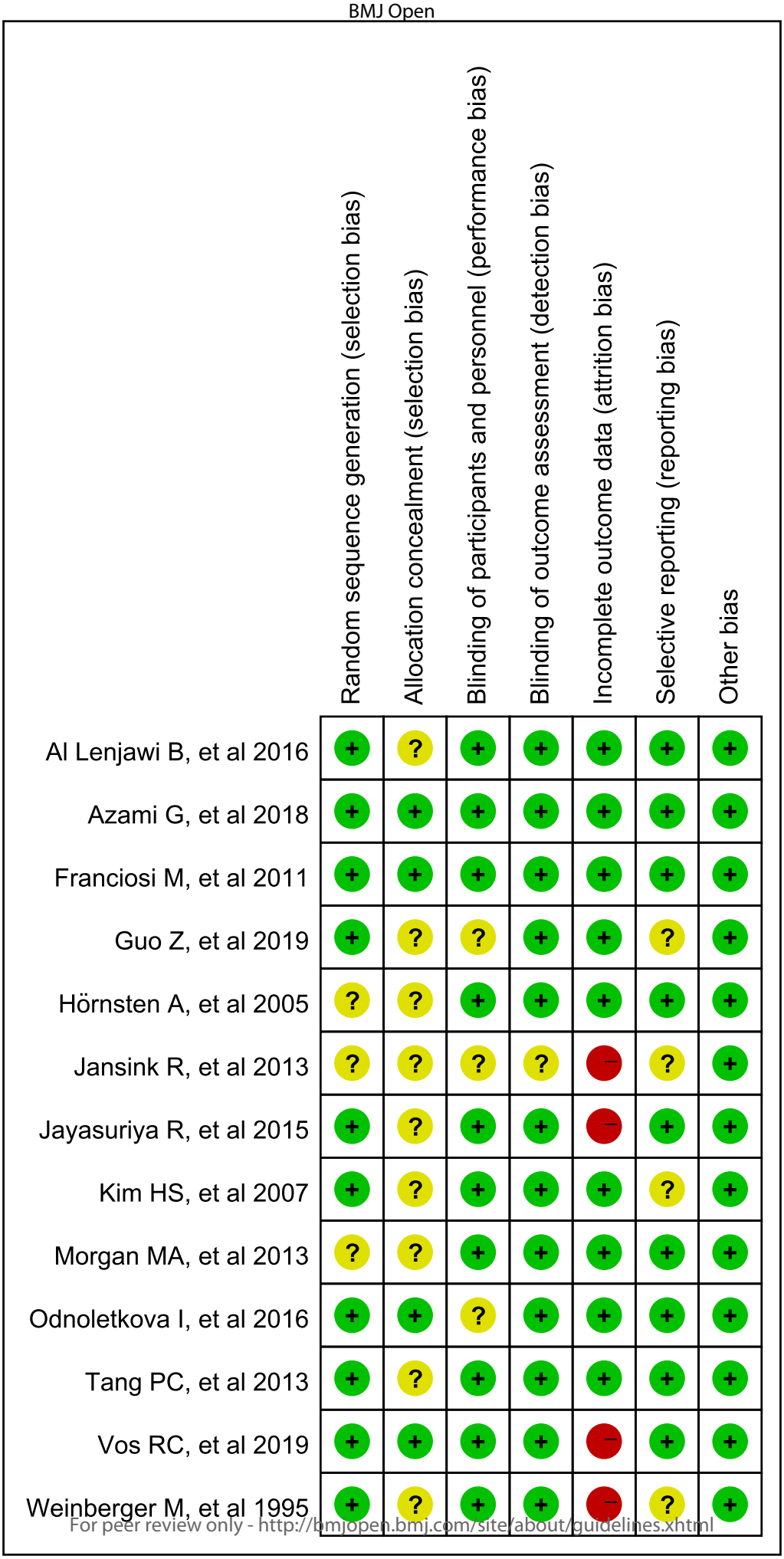
Heterogeneity: Tau² = 11.66; Chi² = 71.27, df = 8 (P < 0.00001); I² = 89%

Test for overall effect: Z = 0.26 (P = 0.79)



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Experimental
Control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
1.1.1 Remote delivery
Kim HS, et al 2007 7.04 1.39 25 7.7 0.9 26 3.7% -0.66 [-1.31, -0.01]
Odnoletkova I, et al 2016 7.4 1 108 7.7 1.2 108 8.6% -0.30 [-0.59, -0.01]
Tang PC, et al 2013 8.1 1.68 186 8.33 1.81 193 7.5% -0.23 [-0.58, 0.12]
Weinberger M, et al 1995 10.5 0.2 188 11.1 0.3 63 12.5% -0.60 [-0.68, -0.52]
Subtotal (95% CI) 507 390 32.3% -0.45 [-0.67, -0.22]
Heterogeneity: Tau² = 0.03; Chi² = 7.43, df = 3 (P = 0.06); I² = 60%
Test for overall effect: Z = 3.92 (P < 0.0001)
1.1.2 Clinial setting
Al Lenjawi B, et al 2016 7.87 1.38 109 8.42 1.99 181 6.8% -0.55 [-0.94, -0.16]
Azami G, et al 2018 7.9 0.93 71 9.3 1.1 71 7.8% -1.40 [-1.74, -1.06]
Franciosi M, et al 2011 6.71 0.1 46 7.2 0.2 16 12.2% -0.49 [-0.59, -0.39]
Jansink R, et al 2013 7.3 0.7 129 7.4 1 197 10.8% -0.10 [-0.28, 0.08]
Jayasuriya R, et al 2015 7 1.2 28 8.3 1.7 25 2.7% -1.30 [-2.10, -0.50]
Subtotal (95% CI) 383 490 40.4% -0.70 [-1.10, -0.30]
Heterogeneity: Tau² = 0.17; Chi² = 49.42, df = 4 (P < 0.00001); I² = 92%
Test for overall effect: Z = 3.46 (P = 0.0005)
1.1.3 Primary care setting
Guo Z, et al 2019 7.18 1 86 7.79 1.33 85 7.5% -0.61 [-0.96, -0.26]
Hörnsten A, et al 2005 5.4 0.7 44 6.4 1.1 60 7.6% -1.00 [-1.35, -0.65]
Morgan MA, et al 2013 6.9 0.26 89 7.4 0.36 67 12.3% -0.50 [-0.60, -0.40]
Subtotal (95% CI) 219 212 27.3% -0.67 [-0.97, -0.38]
Heterogeneity: Tau² = 0.05; Chi² = 7.50, df = 2 (P = 0.02); I² = 73%
Test for overall effect: Z = 4.46 (P < 0.00001)
Total (95% CI) 1109 1092 100.0% -0.58 [-0.73, -0.43]
Heterogeneity: Tau² = 0.04; Chi² = 66.75, df = 11 (P < 0.00001); I² = 84%
Test for overall effect: Z = 7.67 (P < 0.00001)
Test for subgroup differences: Chi² = 1.97, df = 2 (P = 0.37), I² = 0%
-1
-0.5
0
0.5
1
Favours [experimental]
Favours [control]



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118 articles were retrieved by PubMed (search date 2021/4/27)

#1 "diabetes mellitus, type 2"[MeSH Terms] 140199

#2 ((((((((((((((((((((((("Diabetes Mellitus, Noninsulin? Dependent" [Title/Abstract]) OR "Ketosis? Resistant Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Non?Insulin Dependent" [Title/Abstract]) OR "Non？ Insulin？ Dependent Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Stable" [Title/Abstract]) OR "Stable Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Type II" [Title/Abstract]) OR "Diabetes Mellitus, Type 2" [Title/Abstract]) OR "NIDDM" [Title/Abstract]) OR "Diabetes Mellitus, Noninsulin Dependent" [Title/Abstract]) OR "Diabetes Mellitus, Maturity? Onset" [Title/Abstract]) OR "Maturity? Onset Diabetes Mellitus" [Title/Abstract]) OR "MODY" [Title/Abstract]) OR "Diabetes Mellitus, Slow? Onset" [Title/Abstract]) OR "Type 2 Diabetes Mellitus" [Title/Abstract]) OR "Type II Diabetes Mellitus" [Title/Abstract]) OR "Noninsulin? Dependent Diabetes Mellitus" [Title/Abstract]) OR "Maturity? Onset Diabetes" [Title/Abstract]) OR "Diabetes, Maturity? Onset" [Title/Abstract]) OR "Maturity Onset Diabetes" [Title/Abstract]) OR "Type 2 Diabete" [Title/Abstract]) OR "Type II Diabetes" [Title/Abstract]) OR "Diabetes, Type 2" [Title/Abstract]) OR "Diabetes, Type II" [Title/Abstract]) OR "Diabetes Mellitus, Adult? Onset" [Title/Abstract]) OR "Adult? Onset Diabetes Mellitus" [Title/Abstract]) OR "T2DM" [Title/Abstract] 73289

#3 "practice patterns, nurses"[MeSH Terms] 2860

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#4 ((((((((((((((("Clinic?, Nurse? Led" [Title/Abstract]) OR "Nurse? Led Clinic*" [Title/Abstract]) OR "Clinic?,Nurse? Led" [Title/Abstract]) OR "Faculty Practice?, Nursing" [Title/Abstract]) OR "Nurse Practitioner" [Title/Abstract]) OR "Nurse*Role*" [Title/Abstract]) OR "Nurse? Led? Clinic? " [Title/Abstract]) OR "Nurse's Scope of Practice" [Title/Abstract]) OR "Nursing Faculty Practices" [Title/Abstract]) OR "Nursing, Advanced Practice" [Title/Abstract]) OR "Nursing, Private Practice" [Title/Abstract]) OR "Practice Nursing, Advanced" [Title/Abstract]) OR "Practice?, Nursing Faculty" [Title/Abstract]) OR "Practitioner?, Nurse" [Title/Abstract]) OR "Private Duty Nursing" [Title/Abstract]) OR "Private Practice Nursing" [Title/Abstract]) OR "Role?, Nurse*" [Title/Abstract] 7721

#5 #1 OR #2 167151

#6 #3 OR #4 10317

#7 #5 AND #6 118

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1577 articles were retrieved by Cochrane (search date 2021/4/27)

#1 MeSH descriptor: [Diabetes Mellitus, Type 2] explode all trees 18110--18110

#2 (Diabetes Mellitus, Noninsulin? Dependent):ti,ab,kw OR (Diabetes Mellitus, Ketosis? Resistant):ti,ab,kw OR (Ketosis? Resistant Diabetes Mellitus):ti,ab,kw OR (Diabetes Mellitus, Non?Insulin Dependent):ti,ab,kw OR (Non？ Insulin？ Dependent Diabetes Mellitus):ti,ab,kw (Word variations have been searched) 17606--17598

#3 (Diabetes Mellitus, Stable):ti,ab,kw OR (Stable Diabetes Mellitus):ti,ab,kw OR (Diabetes Mellitus, Type II):ti,ab,kw OR (Diabetes Mellitus, Type 2):ti,ab,kw OR (NIDDM):ti,ab,kw (Word variations have been searched) 44169--43950

#4 (Diabetes Mellitus, Noninsulin Dependent):ti,ab,kw OR (Diabetes Mellitus, Maturity? Onset):ti,ab,kw OR (Maturity? Onset Diabetes Mellitus):ti,ab,kw OR (MODY):ti,ab,kw OR (Diabetes Mellitus, Slow? Onset):ti,ab,kw (Word variations have been searched) 849--759

#5 (Slow? Onset Diabetes Mellitus):ti,ab,kw OR (Type 2 Diabetes Mellitus):ti,ab,kw OR (Type II Diabetes Mellitus):ti,ab,kw OR (Noninsulin? Dependent Diabetes Mellitus):ti,ab,kw OR (Maturity? Onset Diabetes):ti,ab,kw (Word variations have been searched) 43213--42971

#6 (Diabetes, Maturity? Onset):ti,ab,kw OR (Maturity Onset Diabetes):ti,ab,kw OR (Type 2 Diabete):ti,ab,kw OR (Type II Diabetes):ti,ab,kw OR (Diabetes, Type 2):ti,ab,kw (Word variations have been searched) 50667--48987

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#7 (Diabetes, Type II):ti,ab,kw OR (Diabetes Mellitus, Adult? Onset):ti,ab,kw OR (Adult? Onset Diabetes Mellitus):ti,ab,kw OR (T2DM):ti,ab,kw (Word variations have been searched) 12871--12537

#8 MeSH descriptor: [Practice Patterns, Nurses'] explode all trees 156--156

#9 (Nurse？Practice Pattern？):ti,ab,kw OR (Practice Pattern?, Nurse's):ti,ab,kw OR (Nurse? Led Clinic*):ti,ab,kw OR (Clinic?, Nurse? Led):ti,ab,kw OR (Nurse? Led Clinic*):ti,ab,kw (Word variations have been searched) 47731--23288

#10 (Clinic?,Nurse? Led):ti,ab,kw OR (Faculty Practice?, Nursing):ti,ab,kw OR (Nurse Practitioner):ti,ab,kw OR (Nurse*Role*):ti,ab,kw OR (Nurse Scope, Practice):ti,ab,kw (Word variations have been searched) 43366--25783

#11 (Nurse？Practice Pattern?):ti,ab,kw OR (Nurse? Led? Clinic?):ti,ab,kw OR (Nurse's Scope of Practice):ti,ab,kw OR (Nurse* Scope, Practice):ti,ab,kw OR (Nursing Faculty Practices):ti,ab,kw (Word variations have been searched) 2370--747

#12 (Practice?, Nursing Faculty):ti,ab,kw OR (Practitioner?, Nurse):ti,ab,kw OR (Private Duty Nursing):ti,ab,kw OR (Private Practice Nursing):ti,ab,kw OR (Role?, Nurse*):ti,ab,kw (Word variations have been searched) 43478--27167

#13 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 54168--52485

#14 #8 OR #9 OR #10 OR #11 OR #12 49120--48217

#15 #13 AND #14in Trials 1577--**1455**

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132 articles were retrieved by embase (search date 2021/4/24)

#1. 'non insulin dependent diabetes mellitus'/exp OR 'non insulin dependent diabetes mellitus'—278657

#2. ('diabetes mellitus' NEAR/3 'noninsulin dependent'):ti,ab—1067

#3. ('diabetes mellitus' NEAR/3 'ketosis resistant'):ti,ab—4

#4. ('diabetes mellitus' NEAR/3 stable):ti,ab—206

#5. ('diabetes' NEAR/3 'type ii'):ti,ab—10

#6. ('diabetes' NEAR/3 'type 2'):ti,ab—210704

#7. 'niddm':ti,ab—8034

#8. ('diabetes' NEAR/3 'maturity onset'):ti,ab—2727

#9. ('diabetes' NEAR/3 'adult onset'):ti,ab—915

#10. ('diabetes' NEAR/3 'slow onset'):ti,ab—27

#11. 'mody':ti,ab—2327

#12. 't2dm':ti,ab—41196

#13. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12—321588

#14. 'nursing practice'/exp—7649

#15. ('practice pattern$' NEAR/3 'nurse*'):ti,ab—61

#16. (clinic$ NEAR/3 'nurs$ led'):ti,ab—1625

#17. ('faculty practice$' NEAR/3 'nurs*'):ti,ab—75

#18. 'nurse$':ti,ab AND near:ti,ab AND 'practitioner':ti,ab—82

#19. 'nurs* role$':ti,ab—4302

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#20. ('nurs* scope' NEAR/3 'practice'):ti,ab—198

#21. ('nurs*' NEAR/3 'advanced practice'):ti,ab—5133

#22. ('nurs*' NEAR/3 'private practice'):ti,ab--121

#23. ('practice$' NEAR/3 'nurs* faculty'):ti,ab—109

#24. ('private duty' NEAR/3 'nursing'):ti,ab—48

#25. ('private practice' NEAR/3 'nursing'):ti,ab—33

#26. ('practice nursing' NEAR/3 'advanced'):ti,ab—1005

#27. ('practice$' NEAR/3 'nursing faculty'):ti,ab—85

#28. #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27—17554

#13 AND #28--132

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524 articles were retrieved by CINAHL (search date 2021/4/25)

#1. MM "Diabetes Mellitus, Type 2"--49719

#2. TI Diabetes Mellitus N3 Noninsulin Dependent--9

#3. TI Diabetes Mellitus N3 Ketosis Resistant--284

#4. TI Diabetes Mellitus N3 Stable--14

#5. TI Diabetes N3 Type II--821

#6. TI Diabetes N3 Type 2--29692

#7. TI NIDDM--709

#8. TI Diabetes N3 Maturity Onset--176

#9. TI Diabetes N3 Adult Onset--80

#10. TI Diabetes N3 Slow Onset--3

#11. TI MODY--146

#12. TI T2DM--366

#13. AB Diabetes Mellitus N3 Noninsulin Dependent--74

#14. AB Diabetes Mellitus N3 Ketosis Resistant--285

#15. AB Diabetes Mellitus N3 Stable--32

#16. AB Diabetes N3 Type II1645

#17. AB Diabetes N3 Type 2--40162

#18. AB NIDDM--1120

#19. AB Diabetes N3 Maturity Onset--402

#20. AB Diabetes N3 Adult Onset--200

#21. AB Diabetes N3 Slow Onset--13

For peer review only

#22. AB MODY--345

#23. AB T2DM--6093

#24. S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23--7104

#25. MM "Nursing Practice+"47366

#26. MH "Nursing Practice+"--69425

#27. MH "Scope of Nursing Practice"--4423

#28. MH "Nursing Practice, Research-Based"--2394

#29. TI Practice Pattern* N3 Nurse*--57

#30. TI clinic* N3 Nurs* Led--520

#31. TI Faculty Practice N3 Nurs*--103

#32. TI Nurse* N1 Practitioner--4395

#33. TI Nurs* Role*--13910

#34. TI Nurs* Scope N3 Practice--519

#35. TI Nurs* N3 Advanced Practice--3253

#36. TI Nurs* N3 Private Practice--69

#37. TI Practice* N3 Nurs* Faculty--113

#38. TI Private Duty N3 Nursing--40

#39. TI Private Practice N3 Nursing--32

#40. TI Practice Nursing N3 Advanced--1512

#41. TI Practice N3 Nursing Faculty--72

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#42. AB Practice Pattern* N3 Nurse*--97

#43. AB clinic* N3 Nurs* Led--817

#44. AB Faculty Practice N3 Nurs*--325

#45. AB Nurse* N1 Practitioner--12118

#46. AB Nurs* Role*--28848

#47. AB Nurs* Scope N3 Practice--880

#48. AB Nurs* N3 Advanced Practice--4845

#49. AB Nurs* N3 Private Practice--94

#50. AB Practice* N3 Nurs* Faculty--315

#51. AB Private Duty N3 Nursing--38

#52. AB Private Practice N3 Nursing--1547

#53. AB Practice Nursing N3 Advanced--19

#54. AB Practice N3 Nursing Faculty--254

#55. S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43 OR S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 --117880

#56. S24 AND S55--524

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	√
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	√
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	9
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	supplemental material
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8-9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8-9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8-9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Fig 1
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	No missing data

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Fig 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Fig 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	eFig 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Fig 2, Fig 3, Fig 4, Fig 5, eFig 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	11-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	11-15
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	11-15
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	11-15
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	11-15
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	11-15
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	16
	23b	Discuss any limitations of the evidence included in the review.	18
	23c	Discuss any limitations of the review processes used.	18
	23d	Discuss implications of the results for practice, policy, and future research.	17-18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	8
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	8
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	none
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	20
Competing interests	26	Declare any competing interests of review authors.	20
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	25-28

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Nurse-led vs. usual-care on cardiovascular risk factors for patients with type 2 diabetes: a systematic review and meta-analysis

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Secondary Subject Heading:	Cardiovascular medicine
Keywords:	DIABETES & ENDOCRINOLOGY, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, EDUCATION & TRAINING (see Medical Education & Training)

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Nurse-led vs. usual-care on cardiovascular risk factors for patients with type 2 diabetes: a systematic review and meta-analysis

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Running Title: Nurse-led care for patients with T2DM

Word count: 3595

Number of tables: 1

Number of figures: 4

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Abstract:

Objects: This study aims to systematically evaluate the effectiveness of nurse-led cares on cardiovascular risk factors among individuals with type 2 diabetes mellitus.

Design: Systematic review and meta-analysis.

Methods: The electronic databases PubMed, EMBASE, CINAHL, and Cochrane Library databases were searched for randomized controlled trials (RCTs) of nurse-led care for individuals with T2DM published in English from inception to December 23, 2021. Random effects models were used to calculate weighted mean differences (WMD) with 95% confidence intervals (95% CI).

Results: 13 articles were included in the meta-analysis, with a total of 3787 participants. Considering baseline measurements, pooled analysis showed that nurse-led care significantly decreased the glycosylated hemoglobin (HbA1c) (WMD = -0.68mmol/L; 95% CI -0.85, -0.52; $p < 0.001$), body mass index (BMI) (WMD=-0.54kg/m^2; 95% CI -0.97, -0.11; $p = 0.01$) and systolic blood pressure (SBP) (WMD = -1.17mmHg; 95% CI -2.11, -0.22; $p = 0.02$) for patients with T2DM. But there was no difference in low-density lipoprotein cholesterol (LDL-c) (WMD=-2.50mmol/L; 95% CI -5.07, 0.08; $p = 0.06$) between the nurse-led and control groups.

Conclusion: Nurse-led care is an effective and accessible intervention that could improve HbA1c, SBP, BMI levels among individuals with T2DM.

PROSPERO registration number: CRD42021248275

Keywords: cardiovascular risk factors; nurse-led care; systematic review; type 2 diabetes mellitus; meta-analysis.

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Strengths and limitations of the study

1 This review provided a comprehensive assessment of the impact of nurse-led care in patients with type 2 diabetes.

2 We use conservative selection methods to minimize erroneous potentially relevant research.

3 Since strict inclusion and exclusion criteria, potential selection bias was minimized.

4 Substantial heterogeneity was observed in these studies.

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1. Introduction

The increasing incidence of type 2 diabetes mellitus (T2DM) has significant implications for healthcare management. According to the World Health Organization (WHO), there are currently more than 463 million T2DM globally[1]. T2DM is associated with several complications[2], in which cardiovascular complications are the main leading of mortality[3-6]. The incidence of myocardial infarction and sudden death in T2DM was the same as in patients with coronary heart disease[7, 8]. Therefore, it is crucial to control the cardiovascular risk factors for patients with T2DM.

Previous studies[9, 10] have shown that most patients with T2DM were under the management of physicians, which has led to a gradual increase in pressure workload for physicians. Hence, many investigators extended the role of nurses as case managers[11]. To address evolving clinical requirements, nurses, may pick a certain number of patients that could be independent within the advanced practice, and thus relieving the stress within medical clinics[12]. There were mounting evidences[13-17] indicating that nurse-led diabetes management programs could provide more comprehensive care, including the guidance of diet, symptom management, lifestyle changes, psychological support, and diabetic education, which could prevent or postpone the complications of diabetes. However, disputable conclusions remained existing in different studies. For example, a study conducted by Tang PC and colleagues[18] showed a great reduction in glycosylated hemoglobin (HbA1c) with an intervention period of 6 months, but no difference for 12 months (P=0.133). Morgan

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MA, et al.[19] found that HbA1c in both experimental groups was remarkable decrease deduced(p=0.049). Vos RC, et al.[20] reported there was no significant difference between the nurse-led and control groups. The inconsistency of these studies has impeded the nurse-led practice in clinical settings. Furthermore, several trials published recently require updated synthesis of evidence. Therefore, we conducted a systematic review and meta-analysis to evaluate the impact of a nurse-led interventions program versus usual-care on patients with T2DM and provide trustworthy practice guidelines on newly released evidence.

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2. Methods

This systematic review was registered on PROSPERO (registering number: CRD42021248275), and conducted following the PRISMA Recommendations project[21]. We devised research questions using the PICO framework for systematic review: for patients with T2DM (Participants), nurse-led diabetes education (Intervention), cardiovascular risk factors outcomes (Outcomes) for the intervention group than the usual care group (Comparison) in randomized controlled trials (RCTs) (Type of studies). We defined nurse-led care[22] as that a nurse played a central role during T2DM management, such as follows up with patients, monitoring blood tests results, and providing continuous education.

2.1 Data sources and searches

The literature search was conducted in PubMed, Embase, CINAHL, and the Cochrane Central Register of Controlled Trials until April 27, 2021. The search strategy was updated on December 23, 2021. We used the following Mesh terms in combination using AND or OR: "Diabetes Mellitus, Type 2", "noninsulin-dependent diabetes mellitus", "Practice Patterns, Nurses", "Nursing Practice". We limited the search language to English. We also searched the reference list of previous reviews related to this systematic review and contacted the study authors for more data when required (The searching strategy is in detailed in supplementary file 1).

2.2 Study selection

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To be included in this review, studies had to meet all the following criteria: (1) randomized controlled trials; (2) had at least one of cardiovascular risk factors for diabetes as an outcome; (3) written in English; (4) and had more than 6 months of follow-up. Trials were excluded (1) if the intervention was delivered by other healthcare providers; (2) duplicate publications.

2.3 Outcomes

The outcome should include any cardiovascular risk factor such as HbA1c, systolic blood pressure (SBP), low-density lipoprotein cholesterol (LDL-c), and Body Mass Index (BMI), or any combination of these.

2.4 Data extraction and quality assessment

Two reviewers (J.Z. and X.Z.) worked independently to scan and evaluate full text for eligible studies according to the criteria of inclusion. The third reviewers (C.L. and Y.D.) resolved disagreements by discussion. Relevant data included: (1) author, publication year, country; (2) delivery setting, age, sample size, length of follow-up; (3) description of interventions, outcomes, and the outcomes with mean and standard deviation checked for accuracy. All data were collated and imported to a statistical evaluation program (Microsoft Excel) and prepared for analysis.

The Cochrane risk-of-bias tool[23, 24] was used to evaluate the bias of randomized studies. Allocation concealment, random sequence generation, blinding of participants,

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personnel and outcome assessors, selective reporting of outcomes, and incomplete outcome data were carefully evaluated.

2.5 Data synthesis and analysis

For the statistical analysis[25], the Review Manager (Version 5.4.0) and Stata MP 16.0. software was used. We have calculated weighted mean differences (WMD) with 95% confidence intervals (95% CI). Heterogeneity was evaluated by $\chi 2$ test and corresponding P-value to assess the dispersion of the actual effect among the included studies. I^2>50% was considered high heterogeneity (random effects model was used), and values lower than 50% to indicate low heterogeneity (fixed effects model was used)[26, 27]. To further explore the effect of different delivery settings, we conducted a subgroup analysis of the factors that may lead to heterogeneity. A sensitivity analysis was performed to explore the effect of each study on the overall pooled estimate. We used funnel plot to detect the publications bias. Trim and fill method will be used to adjust the publication bias if it exists[28]. P value<0.05 indicated a statistically significant difference.

2.6 Patient and Public Involvement statement

The proposed study does not involve patients and the public in the design, or conduct, or reporting, or dissemination plans. Accordingly, no Patient and Public Involvement statement is required.

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3. Results

3.1 Searching process

A total of 2,455 articles were retrieved. After the removal of duplicates, 1,589 articles were retained for abstract and title screening. Initial screening excluded 1,403 articles; only 186 articles remained for full text screening. Further, 173 articles were excluded due to various reasons such as non-nurse-led (n=59), non-eligible population types (n =26), less than 6 months of follow-up (n=18), conference abstract (n=15), non-English articles (n=10), not RCTs (n=13), duplicate reporting (n=9), unrelated to the pre-decided outcomes (n=2), study protocol (n=9), ongoing study(n=4), dissertation (n=1), not relevant (n=1), and no sufficient data (n=6). So, only 13 studies were eventually left for this analysis. The process of screening was displayed in Fig.1.

3.1 Characteristics of the studies included.

Characteristics of included trials and patient characteristics at baseline were shown in Table 1. Of the 13 studies, two were from the United States of America (USA), and the others were from Australia, Netherlands, South Korea, New Zealand, Dutch, Iran, Sri Lank, Northern Sweden, China, Italian, Japan, Belgium, Canada, and the United Kingdom (UK). Care settings and intervention modalities were conducted including remote delivery (Kim HS, et al, 2007; Weinberger M, et al, 1995; Tang PC, et al, 2013; Odnoletkova I, et al, 2016)[18, 29-31], clinical care (Vos RC, et al, 2019; Azami G, et al, 2018; Jayasuriya R, et al, 2015; Franciosi M, et al, 2011; Jansink R, et al, 2013; Al Lenjawi B, et al, 2016)[14, 20, 32-35] and primary care (Hörnsten A, et al,2004; Guo

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Z, et al, 2019; Morgan MA, et al, 2013)[19, 36, 37].

3.2 Nurse-led care program of Included Trials

All studies clearly described programs in nurse-led intervention, such as treatment of case group, intervention methods, practice recruitment, and length of follow-up. The interventions were distributed including structured diabetes education (Hörnsten A, et al, 2005; Al Lenjawi B, et al, 2016; Franciosi M, et al,2011; Jansink R, et al,2013)[33-36], trueBlue collaborative care (Morgan MA, et al, 2013)[19], tele-coaching (Odnoletkova I, et al, 2016; Weinberger M, et al, 1995)[30, 31], online disease management (Tang PC, et al,2013)[18], short message service (SMS) program (Kim HS, et al, 2007)[29], beyond good intentions (BGI) program (Vos RC, et al, 2019)[20] and the diabetes self-management education/diabetes self-management (DSME/DSM) program (Azami G, et al, 2018; Jayasuriya R, et al, 2015; Guo Z, et al,2019)[14, 32, 37]. All studies reported that nurses played a leading role in the entire intervention process. The interventions components were consisted of the face-to-face session, multiple group sessions, and online consulting service. All selected patients received exercise management, blood glucose monitoring, psychological adjustment, drug therapy, and training of insulin injection technology. In two studies (Azami G, et al, 2018; Guo Z, et al,2019)[14, 37], patients received multidisciplinary team management including endocrinologists, nutritionists, nurses, and pharmacists, in which the actual intervention to be carried out by a nurse. In 11 studies, patients received education from different types of nurses (Hörnsten A, et al, 2005; Al Lenjawi B, et al, 2016; Franciosi

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M, et al,2011; Jansink R, et al,2013; Morgan MA, et al, 2013; Odnoletkova I, et al, 2016; Weinberger M, et al, 1995; Tang PC, et al,2013; kim HS, et al, 2007; Vos RC, et al, 2019; Jayasuriya R, et al, 2015)[18-20, 29-36] such as practice nurses (PNs), primary care nurses, nurse care managers (NCMs), and community nurses, et al.

3.3 Methodological quality assessment

The eligible studies included 13 RCTs, and the seven items of e Cochrane Handbook 5.1.0 assessed the quality of these included studies. The methodological quality of studies included was fairly good to moderate, as shown in Fig.2 and Fig.3. Sensitivity analyses would be employed to assess the susceptibility of the findings of this meta-analysis. Which would be carried out by sequential omission of each study and evaluate the effect size changed. We used mention menu in Stata/MP16.0 to conduct sensitivity analyses. The study (Weinberger M)[30] has the most significant impact on the result of meta-analysis. However, its upper limit of 95%CI did not exceed 0. Other studies have no significant changes on the effect size if omitted. This ascertained the robustness of the findings (supplementary file 2).

3.4 Publication bias detection

We used funnel plot to assess the potential publication bias. Asymmetry was found through eyeballing the funnel plot. A trim-and-fill method was applied to estimate and adjust asymmetry in the funnel plot. After filling the right side of the funnel plot, no new studies were imputed and the effect size with confidential interval did not show

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any significant change, which verifies that the conclusion of this study will not be affected by publication bias (supplementary file 3).

3.4 Meta-analysis

3.4.1 HbA1c.

Twelve trials had investigated the effects of nursing intervention on HbA1c. Nurse-led intervention lowered HbA1c compared with usual care (WMD = -0.68mmol/l; 95% CI: -0.85, -0.52; $p < 0.001$) (Fig.4(A)). Meanwhile, we compared the glycation status of the two groups after the intervention, meta-analysis revealed that compared to usual care, nurse-led care also has a positive impact on HbA1c (WMD=-0.58mmol/l, 95% CI = -0.73, -0.43; $P<0.001$) (supplementary file 4).

Subgroup analysis was performed based on different delivery setting (supplementary file 5). All studies in clinical setting (WMD = -0.83mmol/l; 95% CI -1.28, -0.37, $p < 0.001$), primary care setting (WMD = -0.62mmol/l; 95% Cl -1.11, -0.14, $p = 0.01$), and remote delivery (WMD = -0.58mmol/l; 95% CI -0.94, -0.21, $p = 0.002$) showed a greater mean reduction in HbA1c.

3.4.2 BMI.

Nine trials had investigated the effects of nursing intervention on BMI. In the pooled analysis, compared to nurse-led interventions, significant mean difference for BMI was observed for usual care (WMD=-0.54kg/m^2; 95% CI: -0.97, -0.11; $p = 0.01$) (Fig.4(B)).

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However, only two studies were conducted by Franciosi M, et al.[33] and Odnoletkova I, et al.[31] showed a significant difference between the groups over 6 months and 18 months (p = 0.030, p=0.001, p=0.023, respectively). BMI in other studies (Jansink R, et al, 2013; Al Lenjawi B, et al, 2016; Morgan MA, et al, 2013; Vos RC, et al, 2019; Azami G, et al, 2018; Jayasuriya R, et al, 2015; Hörnsten A, et al, 2005;)[14, 19, 20, 32, 34-36] had no significant difference between the groups (p>0.05).

3.4.3 LDL-c.

There was no significant mean reduction in LDL-c in 8 trials for patients in the two group (WMD=-2.50mmol/L; CI 95% -5.07, 0.08; p = 0.06) (Fig.4(C)). and heterogeneity was high (I^2 = 99%, p <0.001). A study by Jayasuriya R, et al.[32] showed that a significant reduction in LDL-c in the "usual care" group but not in the intervention group (p = 0.082). Tang PC, et al.[18] reported that the invention group had significantly better management of LDL-c at 12 months (-6.1mg/dl VS 0.0mg/dl, p=0.001) than that in usual care. Vos RC and colleagues (2019)[20] observed a decrease in LDL-c (p = 0.01) in the intervention group compared to controls. Other studies showed there was no statistically significant difference in LDL-c between the two groups (P>0.05).

3.4.4 SBP.

Of the 13 included studies, 9 investigated SBP levels as a primary outcome in T2DM (Tang PC, et al,2013; Franciosi M, et al,2011; Jansink R, et al, 2013; Odnoletkova I,

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et al,2016; Al Lenjawi B, et al,2016; Morgan MA, et al,2013; Vos RC, et al,2019; Azami G, et al,2018; and Jayasuriya R, et al,2015)[14, 18-20, 31-35]. Significant mean reductions were found for SBP (WMD=-1.17mmHg; 95% CI: -2.11, -0.22; p = 0.02) (Fig.4(D)) in the nurse-led intervention group compared with patients who received usual care. Among these studies, 3 studies (Tang PC, et al,2013; Azami G, et al,2018; and Jayasuriya R, et al,2015)[14, 18, 32] reported that there was a slight improvement form baseline in SBP in the follows-up months in the experimental group when compared to the control group. Other studies (Franciosi M, et al,2011; Odnoletkova I, et al,2016; Morgan MA, et al,2013; Jansink R, et al, 2013; Al Lenjawi B, et al,2016; and Vos RC, et al,2019)[19, 20, 31, 33-35] found that there was no significant difference in SBP between the groups (p>0.05).

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4. Discussion

Nurse-led care is associated with the better protocol compliance, the more regular follow-up, and could provide monitoring of serum chemistry as well as continuous patient education. Regardless of its diverse intervention such as diabetes self-management education[14], SMS[29], telephone counseling[31], or NLTM intervention[37], their common feature are that they all emphasized the leading role of nurses as well as helping patients adhere to treatment plans from different perspectives. This meta-analysis evaluated nurse-led multiple interventions that targeted a wide range of cardiovascular risk factors in patients with T2DM and found that implementing nurse-led care could improve mean HbA1c, BMI, SBP levels in patients with T2DM. However, no improvements were observed in mean differences in LDL-c. The studies we included were from both developed and developing countries, and the participants came from different ethnicities, which may be a good representative worldwide.

HbA1c has been considered a vital element in the therapy of T2DM. Our study found nurse-led care can improve patients' glycemic control, which was consistent with the previous meta-analyses (Daly B et al,2017; Tshiananga JK et al,2012; Wang Q, et al,2019; Yu-Mei Chen D, et al, 2019)[38-41]. The United Kingdom Prospective Diabetes Study(UKPDS) reported that for every 1% lower in HbA1c, microvascular complications would be reduced by 30 percent [42]. In the current study, the overall effect size was 0.68 of a standard deviation improvement in HbA1c compared to usual

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care (Fig.4(A)), which meant nurse-led diabetes care could alleviate glycemic profiles and improve microvascular complications. Subgroup analyses revealed that the effect size of nurse-led programs had positive impacts on glycemic control no matter in different delivery setting. Future research should focus on strategies for sustaining glycemic treatment among nurse-led interventions effects in the long term.

The current study indicated significant weight reduction in the shorter duration (6-12months) nurse-led care was in contrast with previous studies[40, 43]. Obesity and overweight bring a huge financial burden to the individual. In 2014, impact of obesity on the global economic was estimated to be $2.0 trillion of the global gross domestic product[44]. Meanwhile, nurse-led care was found to reduce healthcare costs and be cost-effective[15, 45], which indicates that nurses could greatly reduce the economic burden of patients with higher BMI levels through weight management. Lean et al.[46] had proved that weight management was a lifelong behavior that requires tailored and a strong focus on patient skills training over the patient's lifetime. In this perspective, the role of nurses is particularly important. Nurses participate in patients' treatment, which is the first step in weight loss. Therefore, our meta-analysis is the first we know to emphasize the impact of nurse-led care for at least 6 months duration of achieving remarkable weight loss in patients with T2DM.

Moreover, our results showed a remarkable reduction in SBP compared with usual care, which was consistent with a previous study[39]. For T2DM, higher SBP levels

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are associated with an increased risk of cardiovascular disease. Lower SBP inevitably have a greater impact on physical well-being in patient with T2DM[47]. A previous study report that higher baseline cardiovascular disease risk was associated with greater clinical benefit[48]. When the blood pressure drops to the target range, it is more difficult to continue to lower the blood pressure, so intensive nursing care is needed to help achieve the normal range.

The lack of an effect from nurse-led interventions on LDL-c in our study was in contrast to the finding of Xiao-Dan et al.[43]. It might because nurses in most countries are unable to prescribe and increase titration treatments, and they rely on physicians to play this role, which may hinder the development of nursing roles[49-51]. Improvement of LDL-c needs comprehensive strategies including drugs, diet change, exercise and may not easily be changed by single intervention. Results also suggest that longer-term follow-up education are needed to observe changes in LDL-c[52].

Due to the new developments in the complexity of the disease, T2DM requires long-term care rather than intermittent treatment[53, 54]. This highlights the necessity of nursing work, and nurses can provide guidance and advice to patients in terms of disease management, which could increase the patient's knowledge and self-efficiency. During the special period of the global novel coronavirus pneumonia epidemic, nurses from all over the world are actively responding to the country's call

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in the fight against the epidemic[55]. Before that, nurses perform a neglected human resource in China[56]. In order to improve understanding and respect for the nurse, we need to clearly define the role of nurses. We should also determine the service needs of diabetes nurses and the standards of future careers to reflect the skills and academic achievements of entry and role development.

The limitations of this study are as follows. First, there was significant heterogeneity in all conducted meta-analyses. The cultural and racial differences may be the potential reason for the heterogeneity in 13 trials from 12 countries. Second, compared with other nurse-led RCTs, the sample size of studies we included was small, and the follow-up time of most studies is not sufficient. More high-quality RCTs are needed in the future to help us draw a solid conclusion. Another limitation might be that the follow up time of the 13 studies ranged from 6 months to 30 months. Finally, Type 1 diabetes mellitus is recently increasingly appeared in young adults, more studies are needed to examine this population.

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5. Conclusion

This review demonstrated that nurse-led interventions can improve HbA1c, BMI, SBP levels in patients with T2DM. Nurse-led diabetes education is an indispensable and important part of disease management, which makes patients and their families fully understand the hazards of the disease and increases the awareness of active participation.

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Author Contributions

The study concept and design were framed by CL and YD. JZ and XZ worked independently, scanned the titles, abstracts and evaluated full text for eligible studies using Note-Express software. CL and YD resolved disagreements by discussion. JZ conducted the statistical data analysis and drafted the manuscript. XZ and DM contributed to discussion and revision. CL and YD contributed to revision. All authors read and approved the final manuscript.

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Competing interests

None declared.

Data Availability Statement

All data from systematic review available in paper and supplementary material.

Provenance and peer review

Not commissioned; externally peer reviewed.

Ethics statement

The proposed study does not involve collection of primary data. Accordingly, no ethical approval is required.

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Table Legends

Table 1. Characteristics of included studies.

Figure Legends

Figure 1. PRISMA flow diagram showing study selection.

Figure 2. Risk of bias graph

Figure 3. Risk of bias summary

Figure 4. Comparison of the change scores of cardiovascular risk factors from baseline to follow up

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Study author	Year	Country	Sample size (n)	Delivery setting	follow-up(month)	Mean age of usual care / nurse-led (year)	Intervention team	Treatment of case group	Treatment of control group
Kim HS, et al	2007	South Korea	51	Remote delivery	6	47.5±9.1 /46.8±8.8	Nurse case managers; Physicians	SMS	Usual care
Weinberger M, et al	1995	USA	275	Remote delivery	12	63.2±8.3 /63.9±8.6	Nurse case managers; physicians	Telephone intervention	Usual care
Vos RC, et al	2019	Dutch	108	Clinial setting	30	61.7±7.4 /62.9±8.3	Practice nurses ; physicians	DSME	Usual care
Azami G, et al	2018	Iran	142	Clinial setting	6	53.49±10.98 /55.09±10.16	A multidisciplinary team including endocrinologists, nutritionists, nurses, and pharmacists	DSME+MI	Usual care
Jayasuriya R, et al	2015	Sri Lank	85	Clinial setting	6	51.4±7.1 /51.5±7.5	Nurse case managers; physicians	DSM	Usual care
Hörnsten A, et al	2004	Northern Sweden	104	Primary care setting	12	63.4±9.1 /63.6±9.3	Diabetes nurses; physicians	Diabetes education	Usual care
Guo Z, et al	2019	China	171	Primary care setting	12	64.35±7.07 /63.12±8.02	A multidisciplinary team including community nurses, community doctors,	NLTM	Usual care

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							clinical nursing specialist, diabetes specialists, nutritionist		
Tang PC, et al	2013	USA	415	Remote delivery	12	53.5±10.2 /54.0±10.7	Nurse care managers; pharmacist; registered dietician	EMPOWER-D	Usual care
Franciosi M, et al	2011	Italian	62	Clinial setting	6	48.7±0.6 /48.9±0.5	Diabetes nurses; physicians	disease management	Usual care
Jansink R, et al	2013	Netherlands	940	Clinial setting	14	63.9±9.8 /64.1±8.9	Primary care nurses; physicians	lifestyle counselling	Usual care
Odnoletkova I, et al	2016	Belgium	574	Remote delivery	18	62.4±8.9 /63.8±8.7	Nurse case managers; physicians	COACH	Usual care
Al Lenjawi B, et al	2016	UK	460	Clinial setting	12	55±9.7 /52±8.9	Nurse educators ; physicians	Theory-based educational	Usual care
Morgan MA, et al	2013	Australian	400	Primary care setting	12	67.6±11.2 /68.0±11.7	Practice nurses; physicians	TrueBlue model of collaborative care	Usual care

SMS : short message service; DSME: diabetes self-management education; MI : Motivational Interviewing; DSM: diabete self-management; NLTM : nurse-led team management; EMPOWER-D : Motivating Patients Online With Enhanced Resources for Diabetes; COACH: Tele-education.

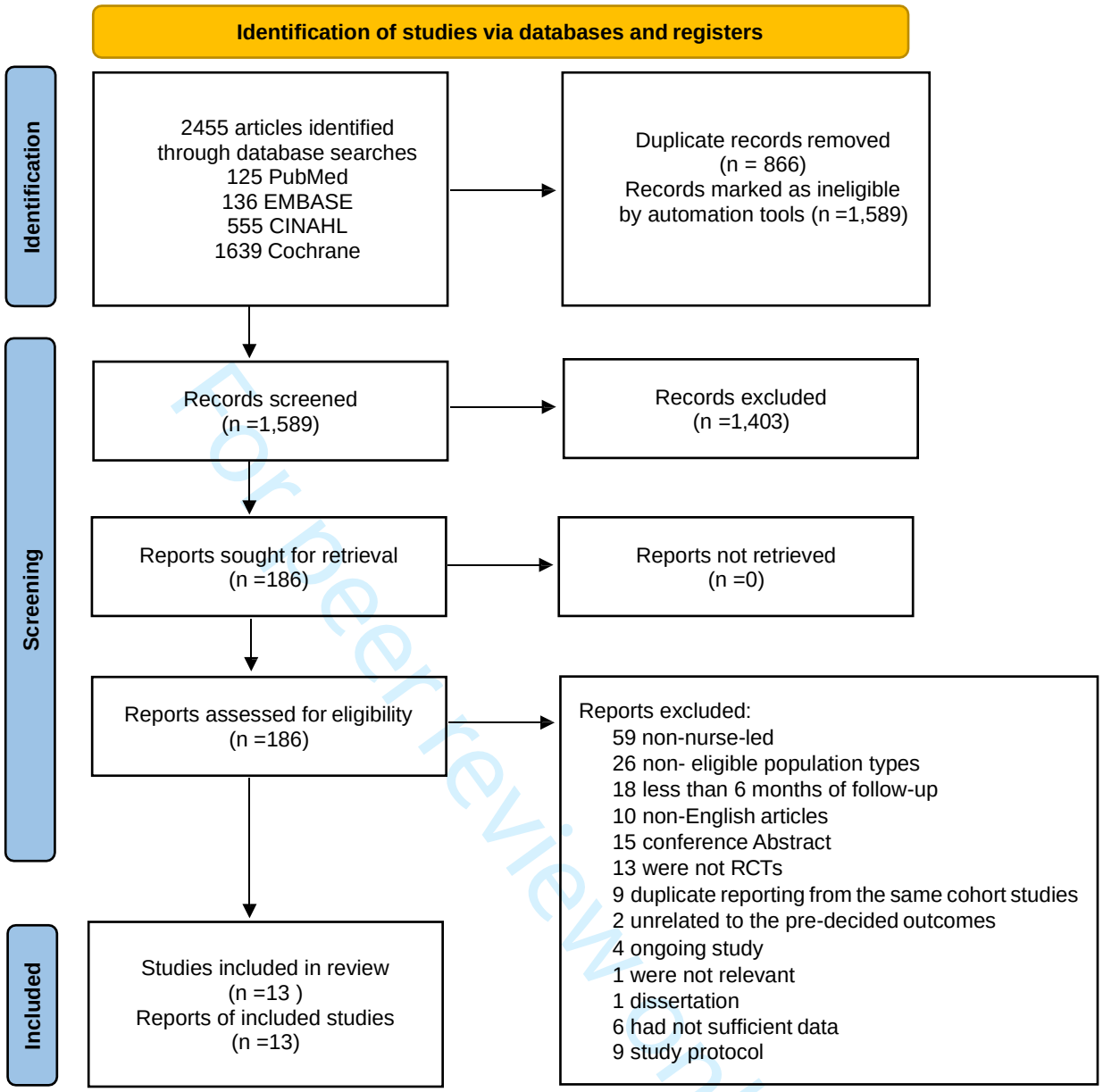
Table.1 Characteristics of included studies

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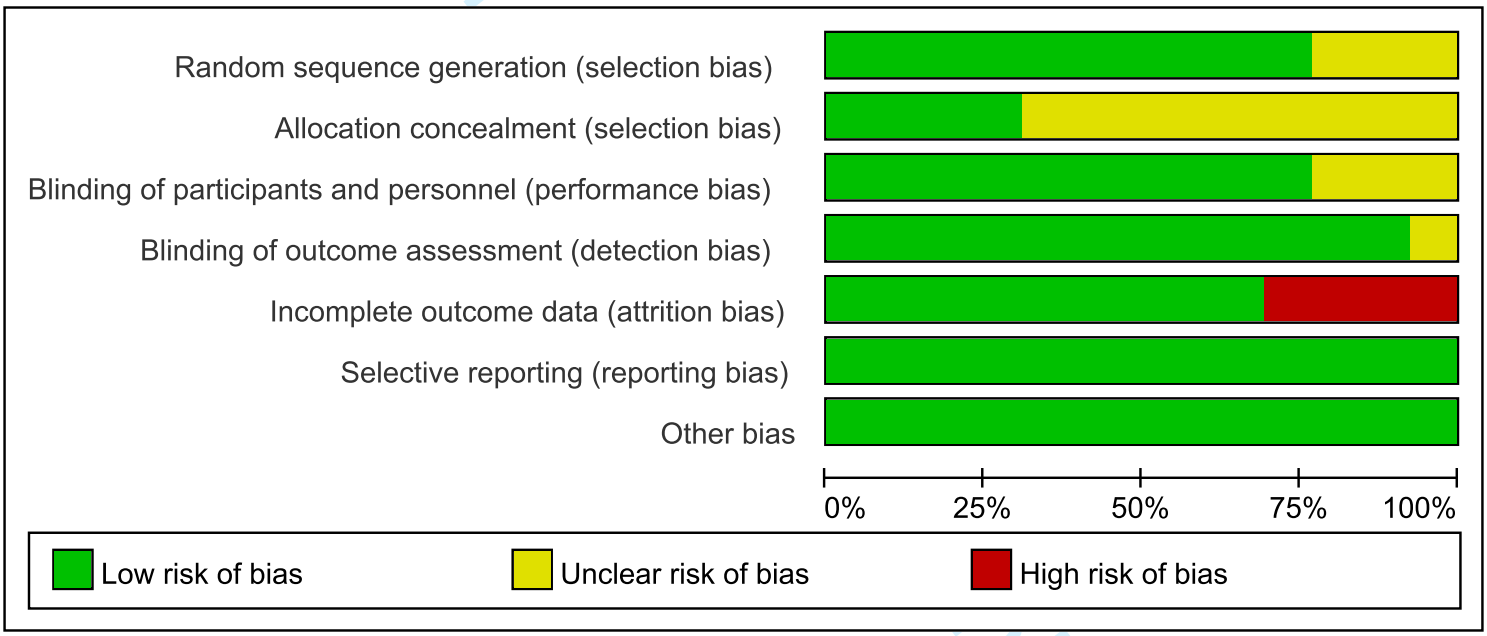
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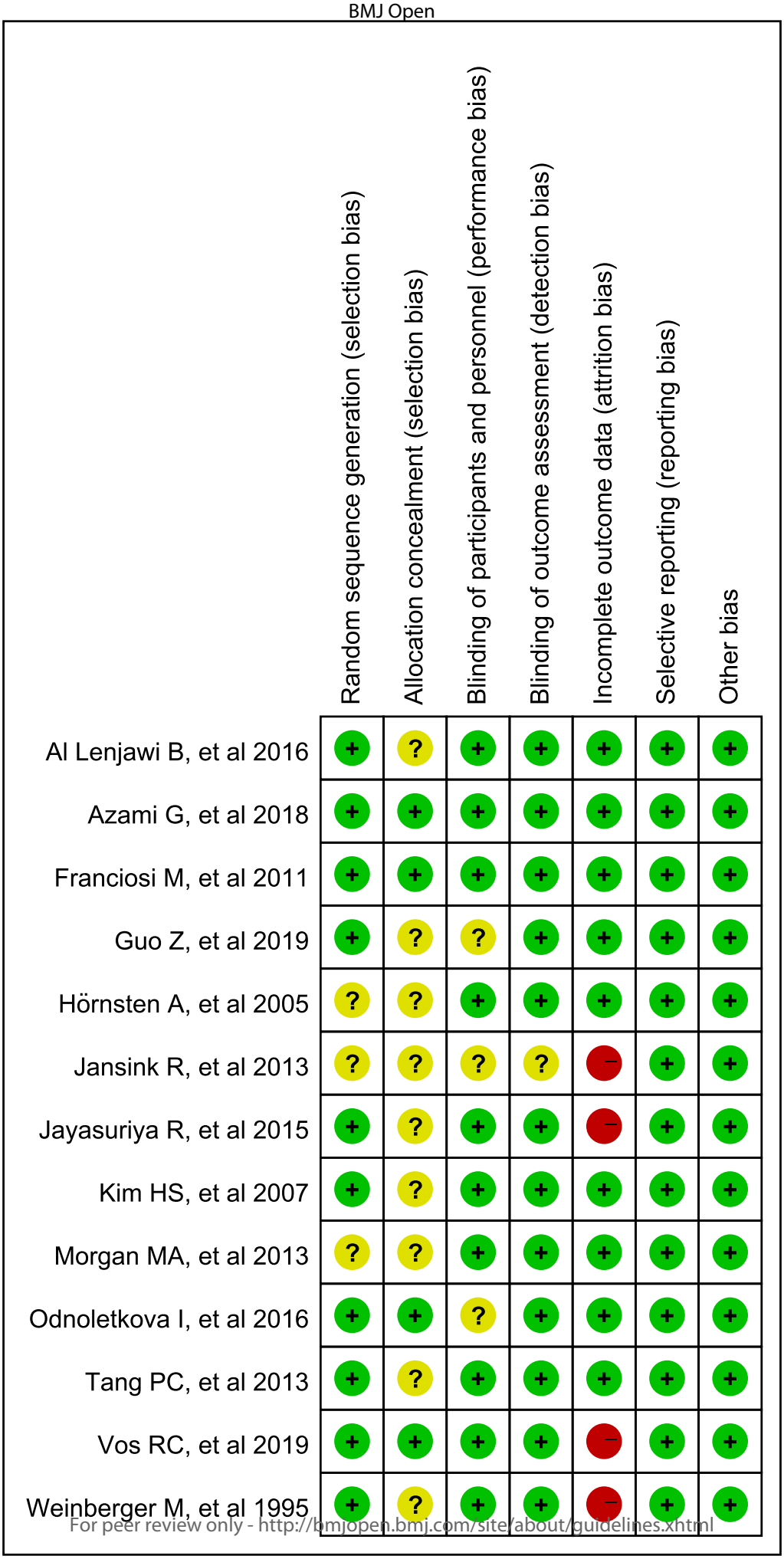
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Fig.1 PRISMA flow diagram showing study selection.



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Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participants and personnel (performance bias)
Blinding of outcome assessment (detection bias)
Incomplete outcome data (attrition bias)
Selective reporting (reporting bias)
Other bias
0%
25%
50%
75%
100%
Low risk of bias
Unclear risk of bias
High risk of bias



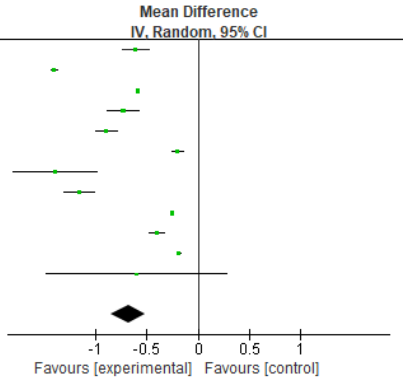
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figure.4 Comparison of the change scores of cardiovascular risk factors from baseline to follow up.

(A)Mean Reduction in HbA1c(mmol/l).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	-0.8	0.12	109	-0.19	0.91	181	8.9%	-0.61 [-0.74, -0.48]
Azami G, et al 2018	-1.42	0.13	71	-0.01	0.05	71	9.3%	-1.41 [-1.44, -1.38]
Franciosi M, et al 2011	-1.29	0.01	46	-0.7	0.01	16	9.4%	-0.59 [-0.60, -0.58]
Guo Z, et al 2019	-0.58	0.71	86	0.15	0.17	85	8.7%	-0.73 [-0.88, -0.58]
Hörnsten A, et al 2005	-0.3	0.1	44	0.6	0.4	60	9.0%	-0.90 [-1.01, -0.79]
Jansink R, et al 2013	-0.5	0.2	129	-0.3	0.3	197	9.3%	-0.20 [-0.25, -0.15]
Jayasuriya R, et al 2015	-2.8	1.1	28	-1.4	0.1	25	6.0%	-1.40 [-1.81, -0.99]
Kim HS, et al 2007	-1.05	0.33	25	0.11	0.19	26	8.7%	-1.16 [-1.31, -1.01]
Morgan MA, et al 2013	-0.07	0.02	89	0.18	0.02	67	9.4%	-0.25 [-0.26, -0.24]
Odnoletkova I, et al 2016	-0.5	0.01	108	-0.1	0.4	108	9.2%	-0.40 [-0.48, -0.32]
Tang PC, et al 2013	-1.14	0.09	186	-0.95	0.07	193	9.4%	-0.19 [-0.21, -0.17]
Weinberger M, et al 1995	-0.2	3.1	188	0.4	3.1	63	2.6%	-0.60 [-1.48, 0.28]
Total (95% CI)			**1109**			**1092**	**100.0%**	**-0.68 [-0.85, -0.52]**

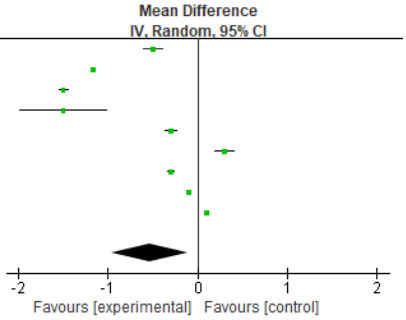
Heterogeneity: Tau² = 0.08; Chi² = 10750.74, df = 11 (P < 0.00001); I² = 100%
Test for overall effect: Z = 7.95 (P < 0.00001)



(B)Mean Reduction in BMI(kg/m2).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	-0.4	0.54	109	0.1	0.03	181	11.2%	-0.50 [-0.60, -0.40]
Azami G, et al 2018	-0.14	0.05	71	1.02	0.06	71	11.3%	-1.16 [-1.18, -1.14]
Franciosi M, et al 2011	-1.6	0.01	46	-0.1	0.1	16	11.3%	-1.50 [-1.55, -1.45]
Hörnsten A, et al 2005	-0.7	0.1	44	0.8	1.9	60	9.9%	-1.50 [-1.98, -1.02]
Jansink R, et al 2013	-0.5	0.2	106	-0.2	0.4	179	11.3%	-0.30 [-0.37, -0.23]
Jayasuriya R, et al 2015	0.3	0.2	28	0	0.2	25	11.2%	0.30 [0.19, 0.41]
Morgan MA, et al 2013	-0.1	0.01	162	0.2	0.2	103	11.3%	-0.30 [-0.34, -0.26]
Odnoletkova I, et al 2016	-0.3	0.1	238	-0.2	0.1	246	11.3%	-0.10 [-0.12, -0.08]
Vos RC, et al 2019	-0.4	0.1	55	-0.5	0.01	51	11.3%	0.10 [0.07, 0.13]
Total (95% CI)			**859**			**932**	**100.0%**	**-0.54 [-0.97, -0.11]**

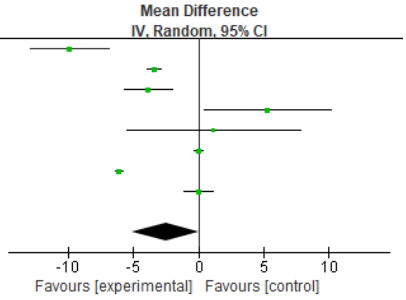
Heterogeneity: Tau² = 0.43; Chi² = 10919.23, df = 8 (P < 0.00001); I² = 100%
Test for overall effect: Z = 2.46 (P = 0.01)



(C)Mean Reduction in LDL-c(mmol/l).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	-17.48	0.1	109	-7.6	20.9	181	12.1%	-9.88 [-12.92, -6.84]
Azami G, et al 2018	-1.94	2.1	71	1.47	0.6	71	14.4%	-3.41 [-3.92, -2.90]
Jansink R, et al 2013	-7.73	7.73	106	-3.87	7.73	178	13.5%	-3.86 [-5.72, -2.00]
Jayasuriya R, et al 2015	-12.9	8.5	28	-18.2	9.5	25	9.5%	5.30 [0.42, 10.18]
Morgan MA, et al 2013	-1.93	23.2	154	-3.09	26.3	86	7.4%	1.16 [-5.50, 7.82]
Odnoletkova I, et al 2016	-10	2	237	-10	2	239	14.5%	0.00 [-0.36, 0.36]
Tang PC, et al 2013	-6.1	1.3	183	0	1.6	189	14.5%	-6.10 [-6.40, -5.80]
Vos RC, et al 2019	-3.87	0	54	-3.87	3.87	50	14.2%	0.00 [-1.07, 1.07]
Total (95% CI)			**942**			**1019**	**100.0%**	**-2.50 [-5.07, 0.08]**

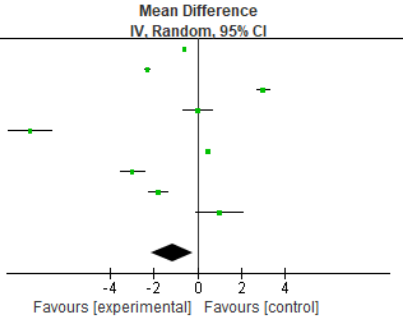
Heterogeneity: Tau² = 11.89; Chi² = 733.57, df = 7 (P < 0.00001); I² = 99%
Test for overall effect: Z = 1.90 (P = 0.06)



(D)Mean Reduction in SBP(mm Hg).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	-1.2	0.17	109	-0.6	0.4	181	11.6%	-0.60 [-0.67, -0.53]
Azami G, et al 2018	-2.9	0.4	71	-0.6	0.3	71	11.6%	-2.30 [-2.42, -2.18]
Franciosi M, et al 2011	-4	1	46	-7	0.01	16	11.5%	3.00 [2.71, 3.29]
Jansink R, et al 2013	-2.9	3.3	120	-2.9	2.2	185	11.0%	0.00 [-0.67, 0.67]
Jayasuriya R, et al 2015	-1.7	2.3	28	6	1.2	25	10.3%	-7.70 [-8.67, -6.73]
Morgan MA, et al 2013	-1.8	0.2	161	-2.3	0.4	112	11.6%	0.50 [0.42, 0.58]
Odnoletkova I, et al 2016	-5	4	241	-2	2	247	11.1%	-3.00 [-3.56, -2.44]
Tang PC, et al 2013	-7.1	3	189	-5.3	1	192	11.3%	-1.80 [-2.25, -1.35]
Vos RC, et al 2019	3	4	55	2	1	51	10.0%	1.00 [-0.09, 2.09]
Total (95% CI)			**1020**			**1080**	**100.0%**	**-1.17 [-2.11, -0.22]**

Heterogeneity: Tau² = 2.01; Chi² = 2412.00, df = 8 (P < 0.00001); I² = 100%
Test for overall effect: Z = 2.41 (P = 0.02)



118 articles were retrieved by PubMed (search date 2021/4/27)

125 articles were retrieved by PubMed (Update 2021/12/23)

#1 "diabetes mellitus, type 2"[MeSH Terms] 140199--150421

#2 ((((((((((((((((((((((("Diabetes Mellitus, Noninsulin? Dependent" [Title/Abstract]) OR "Ketosis? Resistant Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Non?Insulin Dependent" [Title/Abstract]) OR "Non？ Insulin？ Dependent Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Stable" [Title/Abstract]) OR "Stable Diabetes Mellitus" [Title/Abstract]) OR "Diabetes Mellitus, Type II" [Title/Abstract]) OR "Diabetes Mellitus, Type 2" [Title/Abstract]) OR "NIDDM" [Title/Abstract]) OR "Diabetes Mellitus, Noninsulin Dependent" [Title/Abstract]) OR "Diabetes Mellitus, Maturity? Onset" [Title/Abstract]) OR "Maturity? Onset Diabetes Mellitus" [Title/Abstract]) OR "MODY" [Title/Abstract]) OR "Diabetes Mellitus, Slow? Onset" [Title/Abstract]) OR "Type 2 Diabetes Mellitus" [Title/Abstract]) OR "Type II Diabetes Mellitus" [Title/Abstract]) OR "Noninsulin? Dependent Diabetes Mellitus" [Title/Abstract]) OR "Maturity? Onset Diabetes" [Title/Abstract]) OR "Diabetes, Maturity? Onset" [Title/Abstract]) OR "Maturity Onset Diabetes" [Title/Abstract]) OR "Type 2 Diabete" [Title/Abstract]) OR "Type II Diabetes" [Title/Abstract]) OR "Diabetes, Type 2" [Title/Abstract]) OR "Diabetes, Type II" [Title/Abstract]) OR "Diabetes Mellitus, Adult? Onset" [Title/Abstract]) OR "Adult? Onset Diabetes Mellitus" [Title/Abstract]) OR "T2DM" [Title/Abstract] 73289--77665

#3 "practice patterns, nurses"[MeSH Terms] 2860--2939

For peer review only

#4 ((((((((((((((("Clinic?, Nurse? Led" [Title/Abstract]) OR "Nurse? Led Clinic*" [Title/Abstract]) OR "Clinic?,Nurse? Led" [Title/Abstract]) OR "Faculty Practice?, Nursing" [Title/Abstract]) OR "Nurse Practitioner" [Title/Abstract]) OR "Nurse*Role*" [Title/Abstract]) OR "Nurse? Led? Clinic? " [Title/Abstract]) OR "Nurse's Scope of Practice" [Title/Abstract]) OR "Nursing Faculty Practices" [Title/Abstract]) OR "Nursing, Advanced Practice" [Title/Abstract]) OR "Nursing, Private Practice" [Title/Abstract]) OR "Practice Nursing, Advanced" [Title/Abstract]) OR "Practice?, Nursing Faculty" [Title/Abstract]) OR "Practitioner?, Nurse" [Title/Abstract]) OR "Private Duty Nursing" [Title/Abstract]) OR "Private Practice Nursing" [Title/Abstract]) OR "Role?, Nurse*" [Title/Abstract] 7721--8036

#5 #1 OR #2 167151--177638

#6 #3 OR #4 10317--10695

#7 #5 AND #6 118--**125**

For peer review only

1455 articles were retrieved by Cochrane (search date 2021/4/27)

1639 articles were retrieved by Cochrane (Update 2021/12/23)

#1 MeSH descriptor: [Diabetes Mellitus, Type 2] explode all trees 18110--19238

#2 (Diabetes Mellitus, Noninsulin? Dependent):ti,ab,kw OR (Diabetes Mellitus, Ketosis? Resistant):ti,ab,kw OR (Ketosis? Resistant Diabetes Mellitus):ti,ab,kw OR (Diabetes Mellitus, Non?Insulin Dependent):ti,ab,kw OR (Non？ Insulin？ Dependent Diabetes Mellitus):ti,ab,kw (Word variations have been searched) 17606--18777

#3 (Diabetes Mellitus, Stable):ti,ab,kw OR (Stable Diabetes Mellitus):ti,ab,kw OR (Diabetes Mellitus, Type II):ti,ab,kw OR (Diabetes Mellitus, Type 2):ti,ab,kw OR (NIDDM):ti,ab,kw (Word variations have been searched) 44169--43950 46450

#4 (Diabetes Mellitus, Noninsulin Dependent):ti,ab,kw OR (Diabetes Mellitus, Maturity? Onset):ti,ab,kw OR (Maturity? Onset Diabetes Mellitus):ti,ab,kw OR (MODY):ti,ab,kw OR (Diabetes Mellitus, Slow? Onset):ti,ab,kw (Word variations have been searched) 849--874

#5 (Slow? Onset Diabetes Mellitus):ti,ab,kw OR (Type 2 Diabetes Mellitus):ti,ab,kw OR (Type II Diabetes Mellitus):ti,ab,kw OR (Noninsulin? Dependent Diabetes Mellitus):ti,ab,kw OR (Maturity? Onset Diabetes):ti,ab,kw (Word variations have been searched) 43213--45447

#6 (Diabetes, Maturity? Onset):ti,ab,kw OR (Maturity Onset Diabetes):ti,ab,kw OR (Type 2 Diabete):ti,ab,kw OR (Type II Diabetes):ti,ab,kw OR (Diabetes, Type 2):ti,ab,kw (Word variations have been searched) 50667--53194

For peer review only

#7 (Diabetes, Type II):ti,ab,kw OR (Diabetes Mellitus, Adult? Onset):ti,ab,kw OR (Adult? Onset Diabetes Mellitus):ti,ab,kw OR (T2DM):ti,ab,kw (Word variations have been searched) 12871--13571

#8 MeSH descriptor: [Practice Patterns, Nurses'] explode all trees 156--156 163

#9 (Nurse？Practice Pattern？):ti,ab,kw OR (Practice Pattern?, Nurse's):ti,ab,kw OR (Nurse? Led Clinic*):ti,ab,kw OR (Clinic?, Nurse? Led):ti,ab,kw OR (Nurse? Led Clinic*):ti,ab,kw (Word variations have been searched) 47731--50606

#10 (Clinic?,Nurse? Led):ti,ab,kw OR (Faculty Practice?, Nursing):ti,ab,kw OR (Nurse Practitioner):ti,ab,kw OR (Nurse*Role*):ti,ab,kw OR (Nurse Scope, Practice):ti,ab,kw (Word variations have been searched) 43366--45962

#11 (Nurse？Practice Pattern?):ti,ab,kw OR (Nurse? Led? Clinic?):ti,ab,kw OR (Nurse's Scope of Practice):ti,ab,kw OR (Nurse* Scope, Practice):ti,ab,kw OR (Nursing Faculty Practices):ti,ab,kw (Word variations have been searched) 2370--2556

#12 (Practice?, Nursing Faculty):ti,ab,kw OR (Practitioner?, Nurse):ti,ab,kw OR (Private Duty Nursing):ti,ab,kw OR (Private Practice Nursing):ti,ab,kw OR (Role?, Nurse*):ti,ab,kw (Word variations have been searched) 43478--46056

#13 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 54168--56868

#14 #8 OR #9 OR #10 OR #11 OR #12 49120--52070

#15 #13 AND #14in Trials 1455--**1639**

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132 articles were retrieved by embase (search date 2021/4/24)

136 articles were retrieved by embase (Update 2021/12/23)

#1. 'non insulin dependent diabetes mellitus'/exp OR 'non insulin dependent diabetes mellitus'—278657--293744

#2. ('diabetes mellitus' NEAR/3 'noninsulin dependent'):ti,ab—1067--2975

#3. ('diabetes mellitus' NEAR/3 'ketosis resistant'):ti,ab—4--4

#4. ('diabetes mellitus' NEAR/3 stable):ti,ab—206--215

#5. ('diabetes' NEAR/3 'type ii'):ti,ab—10--13

#6. ('diabetes' NEAR/3 'type 2'):ti,ab—210704--222647

#7. 'niddm':ti,ab—8034--8062

#8. ('diabetes' NEAR/3 'maturity onset'):ti,ab—2727--2800

#9. ('diabetes' NEAR/3 'adult onset'):ti,ab—915--946

#10. ('diabetes' NEAR/3 'slow onset'):ti,ab—27--28

#11. 'mody':ti,ab—2327--2419

#12. 't2dm':ti,ab—41196--44372

#13. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12—321588--338658

#14. 'nursing practice'/exp—7649--7757

#15. ('practice pattern$' NEAR/3 'nurse*'):ti,ab—61--63

#16. (clinic$ NEAR/3 'nurs$ led'):ti,ab—1625--1687

#17. ('faculty practice$' NEAR/3 'nurs*'):ti,ab—75--76

#18. 'nurse$':ti,ab AND near:ti,ab AND 'practitioner':ti,ab—82--87

For peer review only

#19. 'nurs* role$':ti,ab—4302--4392

#20. ('nurs* scope' NEAR/3 'practice'):ti,ab—198--212

#21. ('nurs*' NEAR/3 'advanced practice'):ti,ab—5133--5330

#22. ('nurs*' NEAR/3 'private practice'):ti,ab—121--122

#23. ('practice$' NEAR/3 'nurs* faculty'):ti,ab—109--112

#24. ('private duty' NEAR/3 'nursing'):ti,ab—48--48

#25. ('private practice' NEAR/3 'nursing'):ti,ab—33--33

#26. ('practice nursing' NEAR/3 'advanced'):ti,ab—1005--1039

#27. ('practice$' NEAR/3 'nursing faculty'):ti,ab—85--87

#28. #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27—17554--17951

#13 AND #28—132--**136**

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524 articles were retrieved by CINAHL (search date 2021/4/25)

555 articles were retrieved by CINAHL (update 2021/12/23)

#1. MM “Diabetes Mellitus, Type 2”--49719--52456

#2. TI Diabetes Mellitus N3 Noninsulin Dependent--9--9

#3. TI Diabetes Mellitus N3 Ketosis Resistant--284--0

#4. TI Diabetes Mellitus N3 Stable--14--826

#5. TI Diabetes N3 Type II--821--826

#6. TI Diabetes N3 Type 2--29692--32660

#7. TI NIDDM--709--709

#8. TI Diabetes N3 Maturity Onset--176--192

#9. TI Diabetes N3 Adult Onset--80--6

#10. TI Diabetes N3 Slow Onset--3--158

#11. TI MODY--146--167

#12. TI T2DM--366--551

#13. AB Diabetes Mellitus N3 Noninsulin Dependent--74--75

#14. AB Diabetes Mellitus N3 Ketosis Resistant--285--0

#15. AB Diabetes Mellitus N3 Stable--32--34

#16. AB Diabetes N3 Type II1645--1738

#17. AB Diabetes N3 Type 2--40162--43195

#18. AB NIDDM--1120--1125

#19. AB Diabetes N3 Maturity Onset--402--421

#20. AB Diabetes N3 Adult Onset--200--214

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#21. AB Diabetes N3 Slow Onset--13--13

#22. AB MODY--345--365

#23. AB T2DM--6093--6828

#24. S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23--7104--76450

#25. MM "Nursing Practice+"47366--48230

#26. MH "Nursing Practice+"--69425--70767

#27. MH "Scope of Nursing Practice"--4423--4535

#28. MH "Nursing Practice, Research-Based"--2394--2405

#29. TI Practice Pattern* N3 Nurse*--57--60

#30. TI clinic* N3 Nurs* Led--520--590

#31. TI Faculty Practice N3 Nurs*--103--133

#32. TI Nurse* N1 Practitioner--4395--8940

#33. TI Nurs* Role*--13910--14224

#34. TI Nurs* Scope N3 Practice--519--553

#35. TI Nurs* N3 Advanced Practice--3253--3415

#36. TI Nurs* N3 Private Practice--69--73

#37. TI Practice* N3 Nurs* Faculty--113--136

#38. TI Private Duty N3 Nursing--40--40

#39. TI Private Practice N3 Nursing--32--33

#40. TI Practice Nursing N3 Advanced--1512--1564

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#41. TI Practice N3 Nursing Faculty--72--104

#42. AB Practice Pattern* N3 Nurse*--97--101

#43. AB clinic* N3 Nurs* Led--817--863

#44. AB Faculty Practice N3 Nurs*--325--335

#45. AB Nurse* N1 Practitioner--12118--12737

#46. AB Nurs* Role*--28848--29968

#47. AB Nurs* Scope N3 Practice--880--933

#48. AB Nurs* N3 Advanced Practice--4845--5066

#49. AB Nurs* N3 Private Practice--94--95

#50. AB Practice* N3 Nurs* Faculty--315--324

#51. AB Private Duty N3 Nursing--38--19

#52. AB Private Practice N3 Nursing--1547--1615

#53. AB Practice Nursing N3 Advanced--19--38

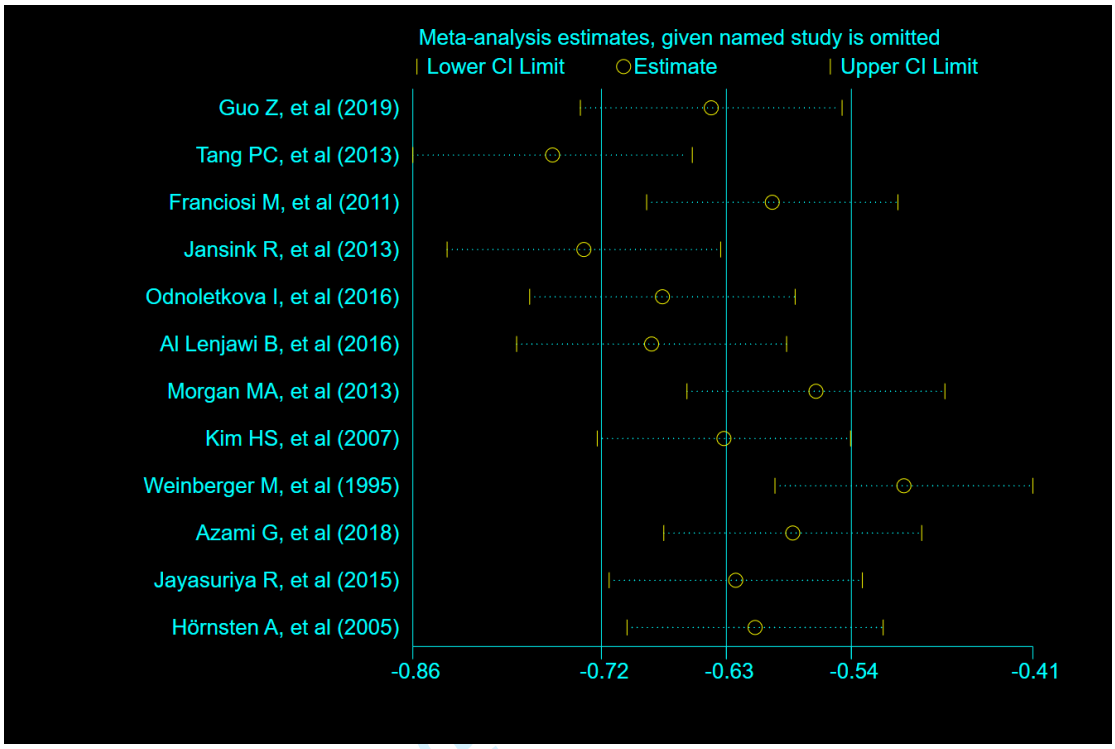
#54. AB Practice N3 Nursing Faculty--254--257

#55. S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43 OR S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54 --117880--122840

#56. S24 AND S55--524--**555**

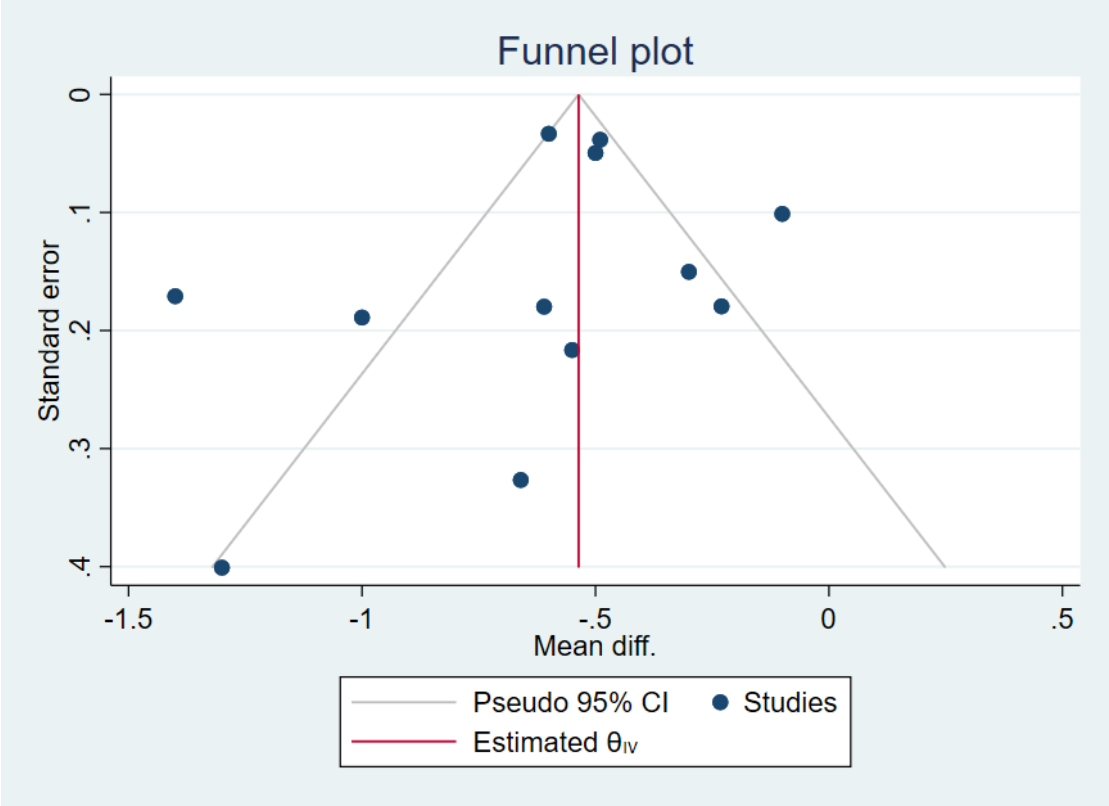
For peer review only

Sensitivity analysis for the studies reporting the outcome 'HbA1c'.

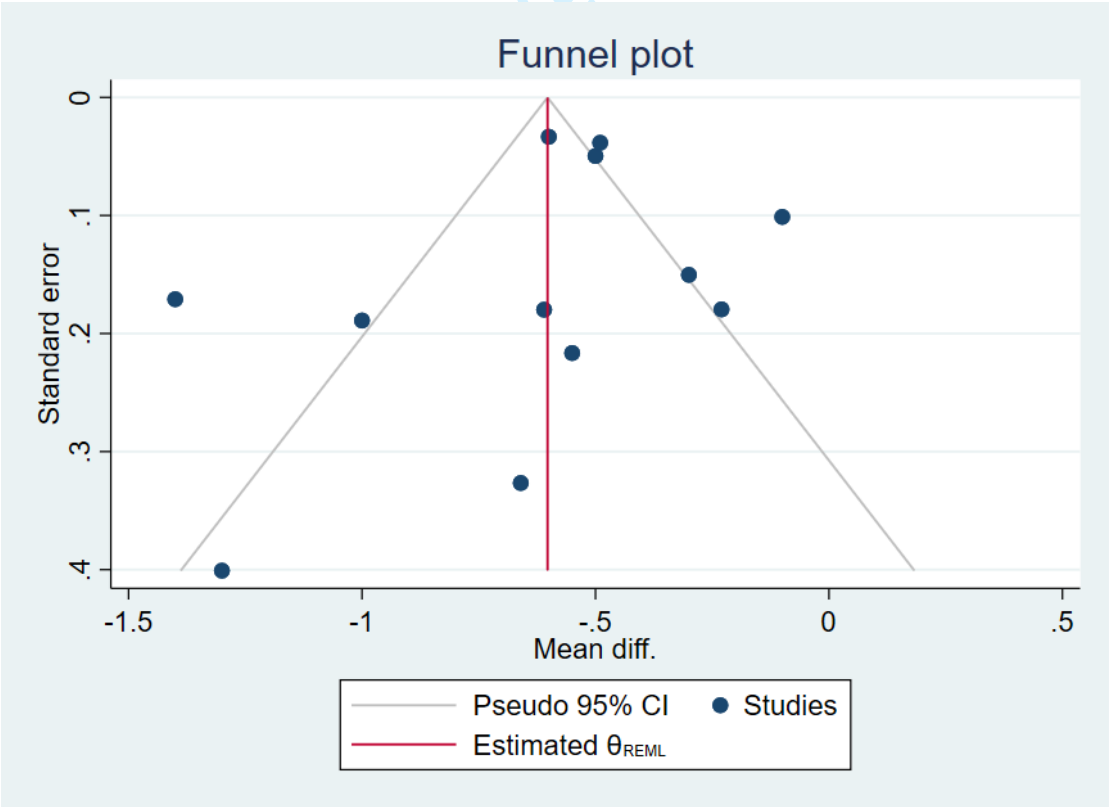


For peer review only

Funnel plot-original



Funnel plot-fill and trim

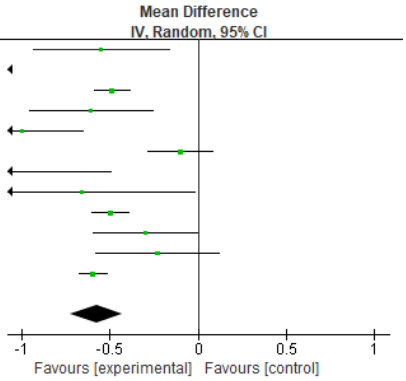


supplementary 4

(A) the forest plot of nurse-led clinic in HbA1c level (mmol/l).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	7.87	1.38	109	8.42	1.99	181	6.8%	-0.55 [-0.94, -0.16]
Azami G, et al 2018	7.9	0.93	71	9.3	1.1	71	7.8%	-1.40 [-1.74, -1.06]
Franciosi M, et al 2011	6.71	0.1	46	7.2	0.2	16	12.2%	-0.49 [-0.59, -0.39]
Guo Z, et al 2019	7.18	1	86	7.79	1.33	85	7.5%	-0.61 [-0.96, -0.26]
Hörnsten A, et al 2005	5.4	0.7	44	6.4	1.1	60	7.6%	-1.00 [-1.35, -0.65]
Jansink R, et al 2013	7.3	0.7	129	7.4	1	197	10.8%	-0.10 [-0.28, 0.08]
Jayasuriya R, et al 2015	7	1.2	28	8.3	1.7	25	2.7%	-1.30 [-2.10, -0.50]
Kim HS, et al 2007	7.04	1.39	25	7.7	0.9	26	3.7%	-0.66 [-1.31, -0.01]
Morgan MA, et al 2013	6.9	0.26	89	7.4	0.36	67	12.3%	-0.50 [-0.60, -0.40]
Odnoletkova I, et al 2016	7.4	1	108	7.7	1.2	108	8.6%	-0.30 [-0.59, -0.01]
Tang PC, et al 2013	8.1	1.68	186	8.33	1.81	193	7.5%	-0.23 [-0.58, 0.12]
Weinberger M, et al 1995	10.5	0.2	188	11.1	0.3	63	12.5%	-0.60 [-0.68, -0.52]
Total (95% CI)			**1109**			**1092**	**100.0%**	**-0.58 [-0.73, -0.43]**

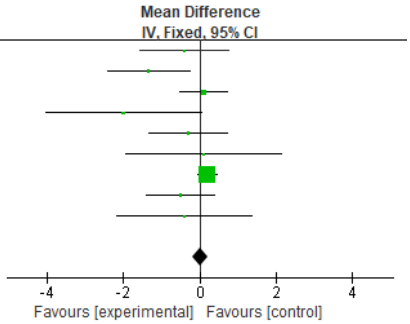
Heterogeneity: Tau² = 0.04; Chi² = 66.75, df = 11 (P < 0.00001); I² = 84%
Test for overall effect: Z = 7.67 (P < 0.00001)



(B) the forest plot of nurse-led clinic in BMI level(kg/m2).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Fixed, 95% CI
Al Lenjawi B, et al 2016	32.4	4.87	109	32.8	4.97	181	3.1%	-0.40 [-1.57, 0.77]
Azami G, et al 2018	28.55	3.2	71	29.89	3.4	71	3.6%	-1.34 [-2.43, -0.25]
Franciosi M, et al 2011	30.2	0.7	46	30.1	1.2	16	11.0%	0.10 [-0.52, 0.72]
Hörnsten A, et al 2005	28.7	4.6	44	30.7	6	60	1.0%	-2.00 [-4.04, 0.04]
Jansink R, et al 2013	30.2	4	106	30.5	4.6	179	4.1%	-0.30 [-1.32, 0.72]
Jayasuriya R, et al 2015	26.5	3.7	28	26.4	3.9	25	1.0%	0.10 [-1.95, 2.15]
Morgan MA, et al 2013	31.2	1	162	31	1	103	69.6%	0.20 [-0.05, 0.45]
Odnoletkova I, et al 2016	29.9	5	238	30.4	5.1	246	5.2%	-0.50 [-1.40, 0.40]
Vos RC, et al 2019	29.2	4.8	55	29.6	4.5	51	1.4%	-0.40 [-2.17, 1.37]
Total (95% CI)			**859**			**932**	**100.0%**	**0.03 [-0.18, 0.23]**

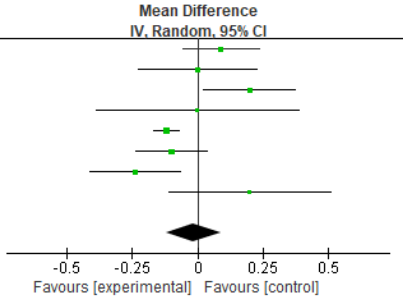
Heterogeneity: Chi² = 14.28, df = 8 (P = 0.07); I² = 44%
Test for overall effect: Z = 0.25 (P = 0.80)



(C) the forest plot of nurse-led clinic in LDL-c level (mmol/l).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	2.98	0.63	109	2.89	0.6	181	14.8%	0.09 [-0.06, 0.24]
Azami G, et al 2018	2.43	0.72	71	2.43	0.65	71	10.7%	0.00 [-0.23, 0.23]
Jansink R, et al 2013	2.6	0.8	106	2.4	0.6	178	13.2%	0.20 [0.02, 0.38]
Jayasuriya R, et al 2015	2.22	0.65	28	2.22	0.78	25	5.4%	0.00 [-0.39, 0.39]
Morgan MA, et al 2013	2.17	0.14	154	2.29	0.2	86	19.9%	-0.12 [-0.17, -0.07]
Odnoletkova I, et al 2016	2.15	0.75	237	2.25	0.78	239	15.3%	-0.10 [-0.24, 0.04]
Tang PC, et al 2013	2.3	0.87	183	2.54	0.84	189	13.3%	-0.24 [-0.41, -0.07]
Vos RC, et al 2019	2.5	0.9	54	2.3	0.7	50	7.5%	0.20 [-0.11, 0.51]
Total (95% CI)			**942**			**1019**	**100.0%**	**-0.02 [-0.12, 0.09]**

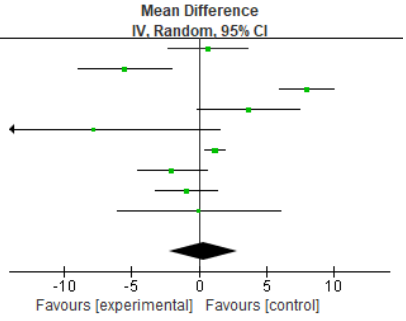
Heterogeneity: Tau² = 0.01; Chi² = 24.68, df = 7 (P = 0.0009); I² = 72%
Test for overall effect: Z = 0.30 (P = 0.76)

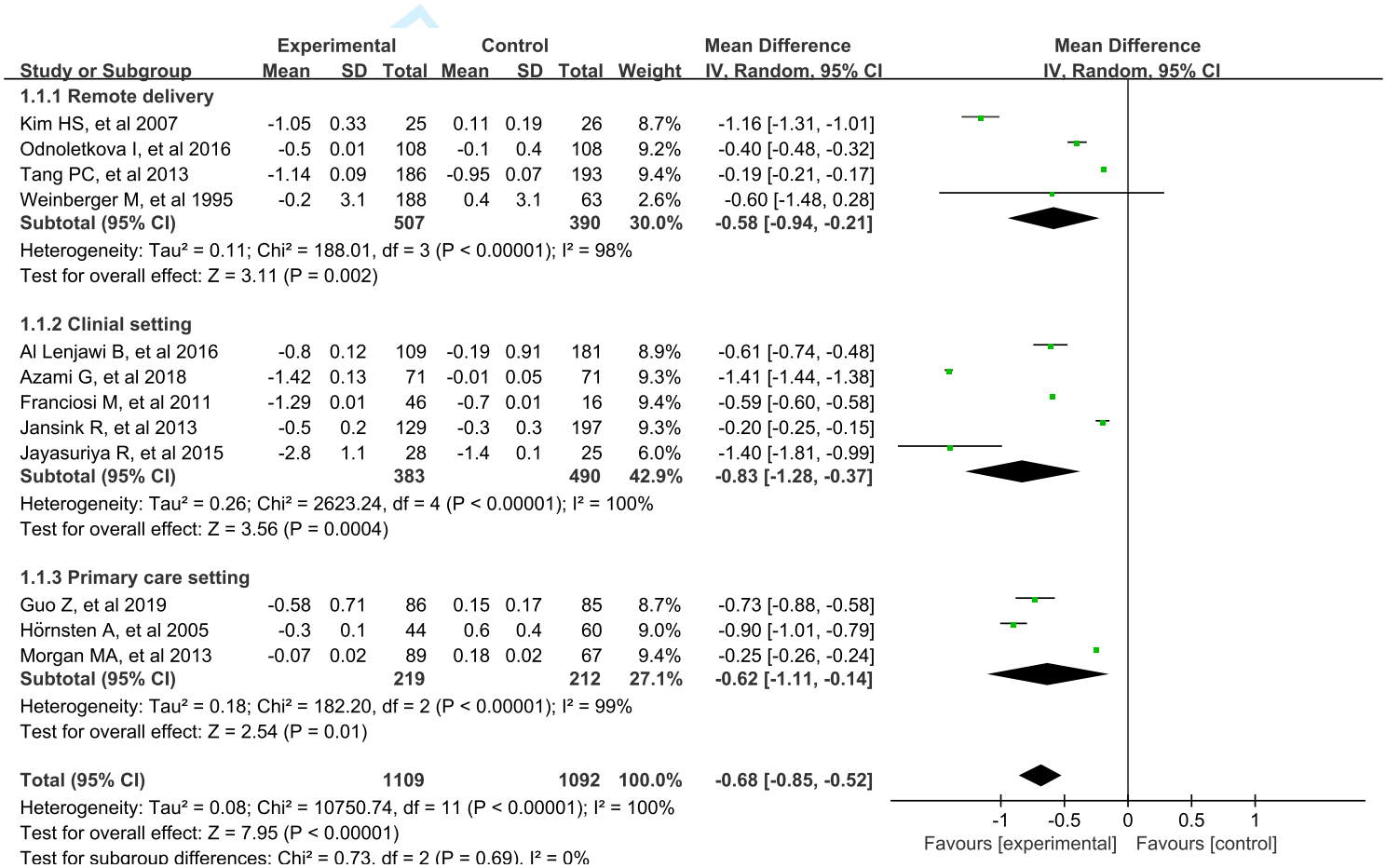


(D) the forest plot of nurse-led clinic in SBP level (mmHg).

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Al Lenjawi B, et al 2016	136.1	12.53	109	135.4	12.3	181	12.1%	0.70 [-2.26, 3.66]
Azami G, et al 2018	127.8	9	71	133.3	12	71	11.3%	-5.50 [-8.99, -2.01]
Franciosi M, et al 2011	133	2	46	125	4	16	13.2%	8.00 [5.96, 10.04]
Jansink R, et al 2013	141.5	17	120	137.8	15.8	185	10.9%	3.70 [-0.10, 7.50]
Jayasuriya R, et al 2015	124.1	15.9	28	131.9	18.6	25	4.9%	-7.80 [-17.17, 1.57]
Morgan MA, et al 2013	132.4	2.8	161	131.2	3.4	112	14.2%	1.20 [0.44, 1.96]
Odnoletkova I, et al 2016	128	14	241	130	15	247	12.6%	-2.00 [-4.57, 0.57]
Tang PC, et al 2013	119.9	11.4	189	120.8	11.5	192	12.9%	-0.90 [-3.20, 1.40]
Vos RC, et al 2019	135	17	55	135	15	51	7.9%	0.00 [-6.09, 6.09]
Total (95% CI)			**1020**			**1080**	**100.0%**	**0.34 [-2.20, 2.89]**

Heterogeneity: Tau² = 11.66; Chi² = 71.27, df = 8 (P < 0.00001); I² = 89%
Test for overall effect: Z = 0.26 (P = 0.79)




Experimental
Control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
1.1.1 Remote delivery
Kim HS, et al 2007 -1.05 0.33 25 0.11 0.19 26 8.7% -1.16 [-1.31, -1.01]
Odnoletkova I, et al 2016 -0.5 0.01 108 -0.1 0.4 108 9.2% -0.40 [-0.48, -0.32]
Tang PC, et al 2013 -1.14 0.09 186 -0.95 0.07 193 9.4% -0.19 [-0.21, -0.17]
Weinberger M, et al 1995 -0.2 3.1 188 0.4 3.1 63 2.6% -0.60 [-1.48, 0.28]
Subtotal (95% CI) 507 390 30.0% -0.58 [-0.94, -0.21]
Heterogeneity: Tau² = 0.11; Chi² = 188.01, df = 3 (P < 0.00001); I² = 98%
Test for overall effect: Z = 3.11 (P = 0.002)
1.1.2 Clinial setting
Al Lenjawi B, et al 2016 -0.8 0.12 109 -0.19 0.91 181 8.9% -0.61 [-0.74, -0.48]
Azami G, et al 2018 -1.42 0.13 71 -0.01 0.05 71 9.3% -1.41 [-1.44, -1.38]
Franciosi M, et al 2011 -1.29 0.01 46 -0.7 0.01 16 9.4% -0.59 [-0.60, -0.58]
Jansink R, et al 2013 -0.5 0.2 129 -0.3 0.3 197 9.3% -0.20 [-0.25, -0.15]
Jayasuriya R, et al 2015 -2.8 1.1 28 -1.4 0.1 25 6.0% -1.40 [-1.81, -0.99]
Subtotal (95% CI) 383 490 42.9% -0.83 [-1.28, -0.37]
Heterogeneity: Tau² = 0.26; Chi² = 2623.24, df = 4 (P < 0.00001); I² = 100%
Test for overall effect: Z = 3.56 (P = 0.0004)
1.1.3 Primary care setting
Guo Z, et al 2019 -0.58 0.71 86 0.15 0.17 85 8.7% -0.73 [-0.88, -0.58]
Hörnsten A, et al 2005 -0.3 0.1 44 0.6 0.4 60 9.0% -0.90 [-1.01, -0.79]
Morgan MA, et al 2013 -0.07 0.02 89 0.18 0.02 67 9.4% -0.25 [-0.26, -0.24]
Subtotal (95% CI) 219 212 27.1% -0.62 [-1.11, -0.14]
Heterogeneity: Tau² = 0.18; Chi² = 182.20, df = 2 (P < 0.00001); I² = 99%
Test for overall effect: Z = 2.54 (P = 0.01)
Total (95% CI) 1109 1092 100.0% -0.68 [-0.85, -0.52]
Heterogeneity: Tau² = 0.08; Chi² = 10750.74, df = 11 (P < 0.00001); I² = 100%
Test for overall effect: Z = 7.95 (P < 0.00001)
Test for subgroup differences: Chi² = 0.73, df = 2 (P = 0.69), I² = 0%
-1
-0.5
0
0.5
1
Favours [experimental]
Favours [control]

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	√
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	√
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	9
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8-9
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	supplemental material
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8-10
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8-10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8-10
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Fig 1
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	10
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	10
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	13

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Fig 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Fig 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Fig 2, Fig 3
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Fig 4
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	14-16
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	14-16
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	14-16
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	14-16
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	14-16
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	14-16
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	17
	23b	Discuss any limitations of the evidence included in the review.	20-21
	23c	Discuss any limitations of the review processes used.	20-21
	23d	Discuss implications of the results for practice, policy, and future research.	17-20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	8
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	8
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	none
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	23
Competing interests	26	Declare any competing interests of review authors.	23
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	23-28

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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